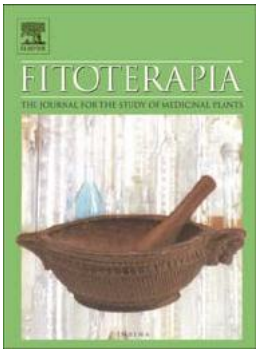

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Traditional uses, fermentation, phytochemistry and pharmacology of *Phellinus linteus*: A review

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Abbreviations:

NMR, nuclear magnetic resonance; ABTS, 2,2'-Azinobis-(3-ethylbenzthiazoline-6-sulphonate); Akt, protein kinase B; Bcl-2, B-cell lymphoma 2; COX-2, cyclooxygenases 2; DC, dendritic cells; DOPA, 3,4-dihydroxyphenylacetic; DPPH, diphenyl-picryl hydrazide; EGFR, epidermal growth factor receptor; FRAP, ferric reducing ability of plasma; GSH-Px, Glutathione peroxidase; IAP, inhibitor of apoptosis protein, IFN, interferon; IL, interleukin; iNOS, inducible nitric oxide synthase; iNOS, inducible nitric oxide synthase; LAK, lymphokine-activated killer; LPO, lipid peroxidation; LPS, lipopolysaccharide; MHC, major histocompatibility complexes; NF-κB, nuclear factor-κB; NK, natural killer cell; ORAC, oxygen radical absorbance capacity; PEG2, prostaglandin E2; PL, *Phellinus linteus*; Reg IV, regeneratinggene IV; ROS, reactive oxygen species; SOD, superoxide dismutase; STZ, streptozotocin; TBRAS, thiobarbituric acid reactive substances; TCM,

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traditional Chinese medicine; TEAC, trolox equivalent antioxidant capacity assay; TLR, toll-like receptors; TNF, tumor necrosis factor; XO, xanthine oxidase.

Abstract

Medicinal mushroom *Phellinus linteus* ("Sanghuang" in Chinese, 桑黄) is a famous fungus which is widely used in China, Korea, and other Asian countries. As a traditional Chinese medicine with a 2000-year long history, medicinal applications of *Phellinus linteus* mainly include treating hemorrhage, hemostasis and diseases related to female menstruation according to Chinese clinical empirical practice. A number of studies reported *Phellinus linteus* possessed good therapeutic effects on various ailments including tumor, diabetes, inflammation, obesity, *etc*. The present paper comprehensively reviewed the traditional uses, fermentation, constituent and pharmacology of *Phellinus linteus* based on scientific literature as well as critical analysis of the research. This review aimed to provide latest information and new foundations and directions for further investigations on *Phellinus linteus*. All available information about *Phellinus linteus* was supplied by library database and electronic search (CNKI, Google Scholar, ScienceDirect, Web of Science, PubMed, *etc*.). Some local and ancient books as well as brilliant scholars were also important information resources. Improvement of fermentation techniques promoted the production of *Phellinus linteus*. Studies of constituents showed the main chemical composition of *Phellinus linteus* included polysaccharides, flavones, triterpenes, aromatic acids, amino acids, *etc*. and polysaccharides were found to account for the largest proportion. Pharmacological researches revealed *Phellinus linteus* possessed a variety of biological activities including anti-cancer, immuno-regulation, anti-diabetes, anti-oxidation and anti-inflammation. Based on these summarized information, this review was presented to provide helpful references and beneficial directions for future studies of *Phellinus linteus*.

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Keywords: *Phellinus linteus*, *Phellinus igniarius*, Traditional Chinese medicine, Fermentation, Phytochemistry, Pharmacology

Chemical compounds studied in this article

Phelligrin A (PubChem CID: 54586497); Naringenin (PubChem CID: 439246); Phelligridins G (PubChem CID: 11444732); Phelligridimer A (PubChem CID: 22835628); Phellinignincisterol A (PubChem CID: 46872463); Phellibaumin A (PubChem CID: 54581651)

1. Introduction

Medicinal mushroom *Phellinus linteus* ('Sanghuang' in Chinese, 'meshimakobu' in Japanese and abbreviated as PL) is a perennial fungus, which belongs to Phellinus Quel., Hymenochaetaceae, Aphyllophorales, Hymenomycetes, Basidiomycotas. Fig. 1 shows medicinal materials of PL and the plant PL grew on. In east Asia, eight taxa were distributed involving *Inonotus baumii* (Pilát) T. Wagner & M. Fisch., *Inonotus lonicericola* (Parmasto) Y. C. Dai, *Inonotus lonicerinus* (Bondartsev) Sheng H. Wu, Y. C. Dai & T. Hatt., *Inonotus sanghuang* Sheng H. Wu, T. Hatt. & Y. C. Dai, *Inonotus vaninii* (Ljub.) T. Wagner & M. Fisch., *Inonotus weigelae* T. Hatt. & S. H. Wu, *Inonotus alpinus* Y. C. Dai & X. M. Tian and *Inonotus zonatus* Y. C. Dai & X. M. Tian [1]. Traditional Chinese medicine (TCM) PL is the dried fruiting body of PL named by parasitizing on the *trunk of Morus alba* L., Morus Moraceae, which was mainly distributed in China, Japan, Mongolia, Korea and some other Asian countries as well as America and Africa. It can also parasitize on other plants like Populus, *Salix babylonica* L., Betula, *etc*. As an excellent and precious medicinal fungus, PL has been widely used as health booster and ancient herbal medicine in East Asian countries, especially in Korea, China and Japan.

PL is a famous medicinal polypore used throughout China, Japan, and Korea with a long history. Its first

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medicinal usage was recorded in an oldest Chinese medicinal book *Shennong's compendium of materia medica* wriiten by 'sanger' (ear of Morus alba L.) in Han dynasty 2000 years ago [2]. Dated back to the beginning of the 20th century, the earliest record of the "meshimakobu" in Japan appeared based on specimens identified as *Fomes yucatanensis* (Murrill) Saccardo & D. Saccardo or *Fomes rimosus* (Berk.) Cooke [3]. The species was named as *Phellinus yucatanensis* (Murrill) Imazeki or *Phellinus linteus* (Berk. & M.A. Curtis) Teng after 1943 [4] and as *Phellinus linteus* in recent decades in Japan [5]. PL, "Sangwhang" in Korean, has been used and prized for its medicinal properties in Korea [6]. *The Characters of Drugs* written by Quan Zhen in the early Tang Dynasty described the medical applications of PL, which was mainly used for hemorrhage, hemostasis and some diseases associated with female menstruation [7]. PL has been also recorded in many other Chinese medical books, as the world's earliest pharmacopoeia issued by the Tang government *New compendium of Materia Medica* [8], the famous *Chinese Compendium of Materia Medica* written by Shi-Zhen Li in the Ming Dynasty [9], and the *Flora Fungorum Sinicorum* [10]. It was an edible fungus with diuresis and hemostasis as well as improvement of internal organs described in *Dictionary of Traditional Chinese Medicine* [11]. *Medicinal Fungus of China* recorded that *Phellinus igniarius*, *Phellinus linteus*, and *Phellinus baumii* were all collectively called "Sanghuang", and it was a rare medicinal fungus exhibiting cytotoxicity, apoptosis, hepatoprotective effect, immuno-regulation and hypoglycemic effect [12]. According to traditional applications and empirical practice, PL has various pharmacological activities including regulating blood glucose, improving blood circulation, hepatoprotecting and enhancing immunologic function, *etc*. Also, PL has been experimentally demonstrated to possess antitumor, immuno-modulatory, anti-inflammatory, anti-oxidant, anti-hyperlipidemic and anti-diabetic activities [13-17].

This review summarized the traditional uses, fermentation, phytochemistry and pharmacology of PL by referring to the published data retrieved from PubMed, Web of Science and Google Scholar, CNKI, *etc*. The current achievements, the next specific research challenges and scientific gaps of PL were analyzed and

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commented as well. The aim of this review is to provide a reference and enlightment for future investigations comprehensively and highlight the importance of PL as well as its utilization and exploitation.

2. Fermentation

Although PL is an excellent medicinal mushroom with a variety of bioactivities and is widely used in East Asia, the wild resources of PL are scarce and limited, which advocates artificial cultivation to improve the production. As thus, fermentation of PL is of great significance and value to promote the exploitation and utilization. To satisfy different popurses and requirements, a large number of recnet studies were conducted to optimize the conditions to find out suitable physiological characteristics for mycelial growth. The results from numerous investigations using Mulberry (*Morus alba*) logs showed that the optimal conditions for mycelial growth of PL were screened as a temperature of 30°C, pH of 6.0 and log length of 20 cm. Optimal culture medium were proposed with supplying sucrose, mannose and glucose as carbon sources and KNO_3 and $NaNO_3$ as nitrogen sources. Sterilized short log inoculation and sterilization over 12 h could accelerate mycelial growth of PL. In addition, burying method of logs after 5-6 months of incubation was beneficial for basidiocarp formation and cultivation house conditions of 31-35°C and more than 96% of relative humidity benefited the formation of fruiting body [18]. Lee et al. [19] reported the maximum mycelial growth rate was 2.80 mm/d when the condition was 29.7 glactose l^{-1}, 26.2°C and pH 5.2 using cheese-processing waste via solid-state cultivation experiments. The production of polysaccharides was used as a reference index to screen suitable fermentation conditions including temperature, pH, inoculum concentration, *etc*. The optimal carbon source and nitrogen source were glucose and soy peptone, respectively. The best inoculation quantity was set at 6% with initial pH of 6.0 at a temperature of 28°C. Rotation speed was set at 180 r/min with adding 0.3 g/L Vitamin B1 and Vitamin B2. Medium volume was set at 60%-65% in shake flask. The best inoculated cell age was 7 days. It was better to add several glass drops into shake flask to promote dispersion, cleavage and proliferation of mycelia [20-22]. Another research compared fermentation and contents of PL inoculated into cooked grains

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(adlay and rice) with that uninoculated controls. It was demonstrated that contents of ash, fat, fiber, total free amino acids, flavor 5'-nucleotides, and protein in inoculated products were much higher than those in uninoculated controls. Total soluble polysaccharides and polyols in PL that fermented into adlay were higher than that into rice, which were all higher than uninoculated controls [23].

Besides the above mentioned physical factors, phytohormones and chemically defined medium was conducive to mycelial growth and polysaccharide biosynthesis [24,25]. By optimizing the chemical ingredients of the medium, the best composition for the production of PL was proposed by Lee et al. [26] as follows: fructose 40 g/L, yeast extract 20 g/L, K_2HPO_4 0.46 g/L, KH_2PO_4 1.00 g/L, $MgSO_4 \cdot 7H_2O$ 0.50 g/L, $FeCl_2 \cdot 6H_2O$ 0.01 g/L, $MnCl_2 \cdot 4H_2O$ 0.036 g/L, $ZnCl_2$ 0.03 g/L and $CuSO_4 \cdot 7H_2O$ 0.005 g/L [26]. However, another reported optimization was 51.67 g/L glucose, 6.96 g/L peptone, 1.0 g/L yeast extract, 5 mg/L thiamine, 1.0 g/L KH_2PO_4, and 0.5 g/L $MgSO_4$ [27,28]. Moreover, supplementation of naphthalentacetic acid, indole-3-acetic acid, NaCl to the submerged cultures of PL at proper concentrations would influence the production of morphology and polysaccharide in PL [24,25,29]. Although the culture conditions of PL have been optimized based on these studies, it is indispensable to select an appropriate one to meet the potential requirements of mass production.

Due to good curative effects on various diseases and scarce wild resources, high price and insufficient production have become a key restraint for PL to increase its medicinal value. Therefore, artificial fermentation for the production of PL is quite momentous. Nevertheless, standard optimized manufacture methods have not been constructed to meet present needs and further studies are needful.

3. Phytochemistry

In the past several decades, a large number of chemical constituents have been isolated and obtained from PL. As is reported, polysaccharides are the main chemical components and primary antitumor active components of PL. Some other classes are also contained including flavones, coumarins, ergosterols, agaricic acids, fatty acids, triterpenes, aromatic acids, amino acids, xylose oxidases, urease, catalases, esterases, sucrases, lactase cellulase

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enzymes, *etc* [30,31]. Polysaccharides, flavone, triterpenes and phenols from PL were still reorted to be responsible for various bioactivities of PL [32,33]. Different trees PL parasitizes on would result in discrepancies in chemical compositions and contents. The contents of polysaccharides, flavonoids and terpenes in PL parasitizing on the *trunk of* Morus alba L. were obviously higher than those on other trees [31]. Additionally, compositions in fruiting bodies and mycelia are also differential. Some chemical components isolated from PL were presented in Table 1. Briefly, polysaccharides, terpenes and steroids are the key components in PL and some endemic compounds have been isolated according to the published materials and they will be subsequently highlighted.

3.1. Polysaccharides

In the fruiting body, polysaccharides are the main components and most of them are heteropolysaccharide. Mannose, galactose, glucose, fructose, xylose are the main monosaccharides in the fruiting body [20]. Most of these polysaccharides were water-soluble. Four polysaccharides typical of PL were found including PIP60-1, PISP1, PL-A and PL-B (Fig. 2). Ge et al. [61] isolated a water-soluble polysaccharide named PBF6 via isolating and purifying by diethylaminoethanol (DEAE)-Sepharose fast flow ion column and Sephacryl S-400 high resolution gel column. Further identification showed the molecular weight of PBF6 was 3.23×10^5 u and the main chain structure of the polysaccharide was →3)-*β*-D-Glc*p*-(1→3)-*β*-D-Glc*p*-(1→6)-*β*-D-Glc*p*-(1→. A water-soluble polysaccharide R1 extracted by 0.1 mol/L NaOH from PL was identified as mannan with main chain 1→6 mannose and side chain 1→3 glucose [62]. Another water-soluble polysaccharide consisted of main chain *β*-(1→3) dextran and side chain *β*-(1→3) mannose which was combined with main chain by *β*-(1→6) glucosidic bond was isolated by Baker et al. [6]. Two heteropolysaccharides named PIP60-1 and PISP1 from the fruiting bodies of *Phellinus igniarius* successively. PIP60-1 was composed of L-fucose, D-glucose, D-mannose, D-galactose and 3-O-Me-D-galactose at a ratio of 1:1:1:2:1 and had a molecular weight of 1.71×10^4 Da. PISP1 was composed of fucose, galactose, mannose and 3-O-Me-galactose at a ratio of 1:2:1:2.

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Methylation and monosaccharide analysis and nuclear magnetic resonance (NMR) studies revealed that PISP1 had a backbone consisted of 1,6-disubstituted-3-O-Me-*α*-D-galactopyranosyl residue, 1, 3, 6-trisubstituted-*α*-D-manopyranosyl residue, 1,4-disubstituted-α-D-galactopyranosyl residue, and had a 1-substituted-*α*-L-fucopyranosyl terminal attached to O-3 of a manopyranosyl residue [41,43]. 3-O-methyl-galactose was firstly obtained and identified by Yang et al. [45] from the fruiting body of PL using ion chromatography, gas chromatography combined with mass spectrometry and NMR.

Two neutral and special heteropolysaccharides, polysaccharide PL-A and proteoglycan PL-B, were separated from the fruiting body and they contained plenty of glucose and few mannose, rhamnose [63]. Ge et al. [38] isolated a polysaccharide, *β*-D-glucan named PBF1, which had a backbone of *β*-(1→6)-linked D-glucopyranosyl residues with a minor of *β*-(1→3)-linked D-glucopyranosyl and terminal *β*-D-glucopyranosyl residues. Another one termed PBF2 containing a *β*-(1→6)-D-glucopyranose backbone with a fucosyl unit on O-3 of the 3, 6-di-O-substituted-D-glucosyl units, a minor (1→3, 6)-*β*-D-mannose residue and terminal glucose residues was obtained previously [44]. Kim et al. [32] isolated an acid proteoglycan from PL by DEAE cellulose anion chromatography method and obtained the molecular weight of the proteoglycan after Sepharose CL-4B gel filtration. Further analysis showed the proteoglycan was consisted of 72.2% polysaccharides and 22.3% protein and the polysaccharide part was mainly composed of mannose, galactose, glucose, arabinose as well as xylose and the protein part contained a lot of glutamic acid, aspartic acid, alanine, glycine and serine. Infrared spectroscopy and ^{1}H and ^{13}C NMR demonstrated that the main chain *β*-(1→3)-D-glucose and the side chain *β*-(1→3)-D-glucose formed the proteoglycan which contained both α-glucosidic bond and *β*-glucosidic [32]. A fucoglucan, PBF4 composed of L-fucose and D-glucose (1:4) was isolated from the fruiting body of PL. Structure information showed that it contained an *α*-(1→4)-D-glucopyranose backbone with some insertions of *α*-(1→2)-L-Fuc*p* residues and a minor *β*-(1→4, 6)-linked D-glucopyranosyl and *β*-glycosidically linked nonreducing-end D-glucopyranosyl residues [35]. Polysaccharides in the fruiting body of PL were

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demonstrated to possess diverse constitutes distinguished from different sources. The primary composition of monosaccharides obtained from Japanese PL was rhamnose, arabinose, glucose, mannose and galactose while monosaccharides involving hamnose, arabinose, xylose, glucose, mannose and galactose were mainly isolated from Korean PL [43].

Compared with the fruiting body, the contents of protein, saturated fatty acids, polysaccharides, reducing sugar, total sugar and amino acids were higher in mycelia [30]. The primary chemical composition in mycelium and zymotic fluid of PL is also polysaccharides, but they are different from the fruiting body. Hwang et al. [34] isolated three exopolysaccharides which were mainly consisted of mannose, galactose and glucose from submerged culture of KCTC 6190 of PL. Lee et al. [71] reported several heteropolysaccharide-protein complexes were obtained from the mycelium of PL. Monosaccharides including glucose, mannose, arabinose and xylose were the main components of the mycelium, and aspartic acid, glutamic acid, glycine and alanine were the minor compounds [71,72]. Song et al. [36] extracted mycelium with hot water and then isolated a polysaccharide PLP, which could stimulate B lymphocyte. It contained 82.5% polysaccharides and 13.2% polypeptides. A β-pyrano polysaccharide HLM consisted of glucose and galactose was isolated by ion exchange and gel chromatography [73].

3.2. Flavones, pyranones and furans

Totally 35 flavones, pyranones and furans have been seperated from PL. Among them, four flavones (phelligrins A and B as well as meshimakobnol A and B (**1-4**)), fifteen pyranones (phelligridimer A, phelligridins A-J, baumin, phellinin C as well as phellinin B1 and B2 (**5-19**)) and five furans (phellinusfrans A and B, phellifuropyranone A, phellinstatin as well as phelliusin A (**20-24**)) were isolated as the characteristic components of PL (Fig. 3). Mo et al. [91] firstly isolated 7 flavones including naringenin, sakuranetin, aromadendrin, folerogenin, eriodictyol coumarin and scopoletin from PL. Afterwards, two new and special ingredients of benzylated dihydroflavones, phelligrins A and phelligrins B, were isolated and identified as 5, 7,

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4'-trihydroxyl-6-O-hydroxylbenzyldihydroflavone and 5, 7, 4'-trihydroxyl-8-O-hydroxylbenzyldihydroflavone [65]. Nagatsu et al. [48] analyzed the methanol extract of fruiting body of PL by ^{1}H detected heteronuclear multiple bond correlation (HMBC) spectroscopic method, and obtained meshimakobnol A and meshimakobnol B.

The polyphenol pigment of PL were mainly pyranones including hispidin skeleton and benzopyrone skeleton. A *β*-secretase inhibitor hispidin ($C_{13}H_{10}O_5$) was isolated from the culture broth of PL [75]. Of pyrone pigment composition, phelligridimer A comprised of highly oxidized hispidin four-polymer and phelligridins A-J was obtained from ethyl acetate fractions of ethanol extraction of *Phellinus igniarius* and phelligridins G, metabolized from 4-hydroxyl-6-methyl-2-pyranones, was a novel skeleton [52,56,80,95]. Lee et al. [57] isolated three new highly oxygenated and unsaturated metabolites from the methanolic extract of the fruiting body and were named as interfungins A-C. In addition, a new styrylpyrone baumin, phellinin B and phellinin C were also obtained and phellinin B was reported as a compound body of phellinin B1 and phellinin B2 [67,92].

Some furans were also reported to exist in PL according to several recent studies. A novel furan-pyrone phellifuropyranone A, a hispidin tripolymer phellinstatin as well as phellinusfrans A and phellinusfrans B were also isolated [51,84,88]. Two stereomeric furan derivatives phellinusfurans A and B, phelliusin A as well as inotilone were also reported [60,93,97].

3.3. Terpenes and steroids

According to some recent reports, some terpenes including sesquiterpenes, ditepenes and triterpenes were seperated from the fruiting body of PL. Eleven terpenes (gilvsins A-D, igniarens A-D and phellilins A-C (**25-35**)) as well as three steroids (phellinignincisterol A-C (**36-38**)) were characterized as the special ingredients of PL (Fig. 4). Wang et al. [66] isolated and identified an abietane diterpene 12-hydroxy-7-oxo-5, 8, 11, 13-tetraene-18, 6-abietanolide and a sterol stigmasta-7, 22-diene-3*β*, 5*α*, 6*α*-triol by spectroscopic methods. In 2009, four lanostane-triterpenoids gilvsins A-D together with two known compounds

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24-methylenelanost-8-ene-3*β*, 22-diol and 5*α*-ergosta-7, 22-diene-3-one were found to exist in the fruiting body of *Phellinus gilvus* [68]. By analyzing the structure information, gilvsins A was demonstrated to be 22-hydroxy-24- methylenelanost-8-en-3-one with acicular crystal, gilvsins B was 3*β*-22R-dihydroxy-24-methylenelanost-8-en-28-oic acid with white amorphous powder, gilvsins C and gilvsins D were 22R-hydroxy-24-methylene-29-norlanost-8-en-3-one and 22R-hydroxy-24-methylene-3-norlanost-8-en-oic acid, respectively. Wang et al. [69] obtained four lanostane-triterpenoids named igniarens A-D (22R -hydroxy-24-methylene-29-norlanost-7, 9(11)-dien-3-one; 3*α*, 22R-dihydroxy-24-methylene-29-norlanost-7, 9(11)-diene; 3*α*, 22R-dihydroxy-24-methylene-29-norlanost-8-ene and 3*α*, 22R-dihydroxy-24-methylenelanost-8-ene, respectively) and they were all amorphous solid. In addition, three novel sesquiterpenes including phellilins A-C were isolated from the ethanol extract of the cultured mycelium of PL [58].

Wu et al. [70] investigated the chemical components of the liquid cultures and mycelium of PL by employing chromatographic methods. Ultimately, four steroids including a homopregnene (3R, 4S, 5S, 17R, 20R-3, 17, 20-trihydroxy-4-methylpregn-8-en-7-one) and three heptanorergosterane derivatives (phellinigncisterol A, phellinigncisterol B, phellinigncisterol C, and nine tremulane sesquiterpenes including (+)-(3S, 6R, 7R)-tremulene-6, 11, 12-triol; (+)-(3S, 6S, 7S, 10S)-tremulene-10, 11, 12-triol; (+)-(3S, 6R, 7R, 10S)-tremulene-6, 10, 12-triol; (-)-(2S, 3S, 6S, 7S, 9R)-tremul-1(10)-ene-11, 12, 14-triol; (-)-(2S, 3S, 6S, 7S, 9S)-tremul-1(10)-ene-11, 12, 15-triol; (-)-(2R, 3S, 6S, 7S, 9R)-tremul-1(10)-ene-11, 12, 14-triol; (-)-(2S, 3S, 4S, 6S, 7S)-tremul-1(10)-ene-4, 11, 12-triol; (+)-(2S, 3R, 6S, 7S)-tremul-1(10)-ene-2, 12-diol; (+)-(1R, 6S, 7S)-tremul-2-ene-12(11)-lactone as well as some other known compounds were isolated from the cultures of *Phellinus igniarius* [47,70,71]). Additionally, three sesquiterpenes including (3S, 9R, 10S-3*β*-hydroxy-11, 12-O-isopropyldrimene, 3S, 9R, 10S-3*β*, 11, 12-trihydroxydrimene, 3S, 4S, 9R, 10S-11, 12, 14-trihydroxydrimene) and three steriods including ergosta-4, 6, 8(14), 22-tetraen-3-one, stigmasta-7,

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22-diene-3*β*, 5*α*, 6*α*-triol, 5*α*, 8*α*-epidioxyergosta-6, 22-diene-3*β*-ol were purified by Silica gel column chromatography, Sephadex LH-20 column chromatography and reversed-phase HPLC column chromatography *etc.* and their structures were further confirmed by NMR spectroscopy and mass spectrometry [74].

3.4. Other components

In addition to above components, a range of other compounds were also obtained from PL. Twelve compounds typical were separated and identified in PL including phellinins A1, A2 phellibaumin A-E, methyl-phelligrins A and B as well as isomethyl-phelligrins A and B (**39-50**) (Fig. 5). Samchai et al. [59] studied the dichloromethane fraction of the PL and elucidated two indole derivatives including 7-methoxyindole-3-carboxylic acid methyl ester, 1-methylindole-3-carboxaldehyde and a phenolic compound (E)-4-(3, 4-dihydroxyphenyl)but-3-en-2-one based on NMR spectroscopic analysis and mass spectrometric data. Two new styrylpyrones phellinins A1 and A2 were isolated from the cultured broth of *Phellinus* sp. KACC93057P [53]. Wu et al. [74] firstly isolated and identified 11 novel compounds, and named them as phellibaumin A-E, methyl-phelligrins A and B, isomethyl-phelligrins A and B, (22E, 24R)-6*β*-oxethyl-ergosta-7, 22-diene-3*β*, 5*α*-diol and (22E, 24S)-6*β*-oxethyl-ergosta-7-ene-3*β*,5*α*-diol from agrestal *Phellinus baumin* Pilát. Besides the above-mentioned compounds, other minor compounds were discovered successively. Six simple secondary metabolites including methyl-4-hydroxyphenylacetate, salicylaldehyde, 2, 5-bis (hydroxymethyl) furan, succinic acid, 2-methylol-5-methoxylfuran and N-acetyl-2-p-hydroxyphenylethylamine from PL [90]. Cluytyl ferulate, eugenol, protocatechualdehyde, protocatechuic acid, caffeic acid, iso-ergosteion, microelements [70], phellinone [54], cyclo-dipeptide, amino acids [74], *etc* were further found in PL.

4. Pharmacology

4.1. Antitumor activities

In general, antitumor is recognized as the most important bioactivity of PL and PL is verified as an effective anti-cancer medicine by lots of pharmacological experimental studies. In 1968, Ikekaw et al. [100] firstly

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discovered that the extract from the fruiting bodies of PL could obviously inhibit mouse sarcoma cells (S180) at a rate of 96.7%. Numerous investigations indicated that polysaccharides were the main components to exert antitumor effect in PL [101]. Besides, other extracts of PL also significantly prevented cancers based on recently published data. The antitumor activities of PL were summarized and listed in Table 2.

4.1.1. Antihepatocarcinoma activitiy

Extensive researches demonstrated anti-hepatocarcinoma effect was one of the most significant antitumor activity. Polysaccharides from PL could make a good performance for inhibiting HepG2 human hepatocellular carcinoma cells. The results showed that the polysaccharides with a concentration of 0-1.0 mg/mL markedly suppressed the proliferation of HepG2 cells by 50% in a dose- and time-dependent manner [109]. Furthermore, the inhibitory activity was exhibited via inducing S-phase cell cycle arrest at a concentration of 1.0 mg/mL and prevented cells from forming colonies, but PL induced the formation of apoptotic morphology in HepG2 cells , which was associated with a reduction in B-cell lymphoma 2 (Bcl-2) level and an increase in the release of cytochrome c when the concentration was increased to 1.5 and 3 mg/mL [109]. From these results, PL was suggested to have a direct antitumor effect by initiating apoptosis and cell cycle blockade in HepG2 cells. The antitumor effect of polysaccharides extracted from PL on Hep2 cells was also studied and confirmed. With the treatment of 200 mg/kg, the volume and weight of solid tumors were significantly decreased compared with control mice [107]. Moreover, 200 μg/mL of PL polysaccharides extract caused a significant dose-dependent increase in S phase arrest, and obviously down-regulated expressions of calreticulin, cyclin D1, cyclin E, and CDK2 and up-regulated expressions of P27kip1 and cyclin A in treated HepG2 cells [107]. Xue et al. [105] isolated a water-soluble polysaccharide from mycelia of PL which exerted immunoregulatory and antitumor activities *in vitro*. The results indicated that the proliferation of HepG2 human liver cancer cells could be inhibited by inducing cell cycle arrest at S phase at a concentration of 400 μg/mL, which led to apoptosis [105]. Treatment of endo-polysaccharide from *Phellinus igniarius* significantly elevated serum IL-2 and IL-18 in H22

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and S180 implanted mice at the doses of 500 mg/kg and 250 mg/kg [110]. The outcome indicated *Phellinus igniarius* exert an antitumor effect through enhancement of cell mediated immunity. A study was conducted to reveal the anti-proliferative and anti-metastatic activity of the ethanolic extract from the fruiting body of *Phellinus igniarius* on human hepatocarcinoma SK-Hep-1 cells and rat heart vascular endothelial cells. With 25 μg/mL of the extract, the secretion of matrix metalloproteinase-2 and vascular endothelial growth factor were significantly decreased in SK-Hep-1 cells and the formation of matrigel-induced tube was inhibited in rat heart vascular endothelial cells [102]. More importantly, 25 or 50 μg/mL of the extract combined with oxaliplatin (Oxa) or 5-flurouracil (5-FU) synergistically blocked the proliferation of SK-Hep-1 cells [102]. Hispolon of PL was also found to perform an anti-proliferative effect on human hepatocellular carcinoma Hep3B cells. According to a previous study, hisplolon inhibited cellular growth through the induction of cell cycle arrest at S phase in a time-dependent and dose-dependent manner, which was associated with a remarkable down-regulation in expressions of cyclins A and E and cyclin-dependent kinases 2, with concomitant induction of p21waf1/Cip1 and p27Kip1 [104].

Many other investigations were also conducted to elucidate and discuss the mechanism on the anti-hepatocarcinoma of different active compounds obtained from PL [103,106,108,151-153].

4.1.2. Antimelanoma activitiy

PL not only inhibited growth of liver cancer but infected melanoma cells. The extract of PL (40, 200 and 1000 g/kg/d) was orally gavaged to C57BL/6 mice after or before implanting H22 (hepatocellular carcinoma) and B16 (melanoma) cells and the anticancer effect was then assessed. The results showed cytokines, IL-12 and TNF-α in H22 cells and B16 cells treated with PL were higher than untreated group and the levels of $CD4^+$ T cell and $CD4^+/CD8^+$ had a positive correlation with the doses in B16 cells but inhibitory activity of LAK cell presented a negative correlation with the doses in H22 cells and it was reversed to that in prevention of B16. Inhibitive activity of NK cells in high-dose groups was stronger than that in other groups in B16 but it was

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similar among all the three groups in H22. By performing a H22 treated experiment, the extract was oral administrated and the indexes of immune organs thymus and spleen were markedly increased [111, 112]. These results indicated that both progression of tumor and metastasis of tumor cells could be prevented by PL. Meanwhile, PL regulated the growth of liver tumor and melanoma cells in different ways, whch was consistent with the differences of biochemical indicators between hepatocellular carcinoma and melanoma cells.

Studies on the anti-melanoma activity of PL were conducted and reported by some investigators. Acidic polysaccharide from PL (100 μg/mL) directly inhibited cancer cell adhesion and strongly reduced B16F10 murine melanoma cells invasion through extracellular matrix, but no direct effect on growth of cancer cells was found [116]. Ohno et al. [150] discovered that dietary intake of the powder of PL fruiting body (83, 415 and 2075 mg/kg/d) notably prolonged the survival time of B16/BL6-bearing mice and ingestion of PL dose-dependently enhanced NO production by macrophages prepared from the tumor-bearing [150].

The inhibitory effect of PL on B16BL6 cells was additional evaluated. Park et al. [114] investigated antitumor activity of the ethyl acetate extract of PL parasitizing on *Panax ginseng* and the reults indicated pronounced anti-proliferative activities were exhibited at the concentrations of 1, 10, 100 and 500 μg/mL through inducing apoptosis and cellular differentiation, which increased a sub-G1 cell population by inducing p53/p21 and activated caspase-8 protein expression in B16F10 cells. Furthermore, Lee et al. [115] found that the aqueous extract of Cambodian PL did not only significantly induce cytotoxicity against B16BL6 cells (125, 250, 500, 1000 μg/mL) but also inhibited the pulmonary metastatic colonies in C57BL/6 mice intravenously injected by B16BL6 cells when oral administration was given at the dose of 2, 20, and 50 mg/kg. Also, down-regulated expression of urokinase type plasminogen activator was observed. In addition, Han et al. [113] assessed the anti-melanoma effect of polysaccharides isolated from PL and held the point that its bioactivity was due to immune-enhancement [113].

4.1.3. Anti-colorectal carcinoma activity

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To study anti-colorectal carcinoma activity, several kinds of colon cancer cells were generally used including SW480, HT-29 as well as HCT116 cells, *etc*. Protein-bound polysaccharide from PL with concentrations of 125-500 mg/mL inhibited the proliferation and colony formation of SW480 human colon cancer cells by reducing cyclin B1 and B-cell lymphoma-2, but it inreased the release of cytochrome c and the populations of both apoptotic sub-G1 and G2/M phase [117]. Polysaccharides extracted from PL (125-1000 μg/mL) could also inhibit cell proliferation and decrease *β*-catenin expression in SW480 cells. The possible mechanism was that it reduced the expression of downstream-regulated genes *β*-catenin cyclin D1 and inhibited T-cell factor/lymphocyte enhancer binding factor transcription activity as well as matrix metallo-proteinase activity [118]. The antitumor study of polysaccharides from PL was performed on human colorectal carcinoma HT-29 cells, colorectal carcinoma (Caco-2) cells and human hepatocyte (LO-2) cells *in vitro* and *vivo*. *In vivo*, the growth of HT-29 cells were significantly restrained in tumor-challenged female BALB/c-nu/nu mice with the oral administration of polysaccharides at the doses of 100 and 200 mg/kg/d [119]. With a nontoxic dose of polysaccharides, a decrease of volume and weight of tumor cells was observed in HT-29-bearing mice. *In vitro*, the polysaccharides increased S-phase cell cycle in a significantly dose-dependent manner but no apoptosis and autophagy in HT-29 cells. Furthermore, the polysaccharides (100 μg/mL) notably down-regulated expressions of cyclin D1, cyclin E and CDK2 in HT-29 cells, while the expression of P27kip1 was down-regulated [101]. Proteoglycan (100 mg/kg, 15.625-250 μg/mL), purified from PL, relatively increased weights of spleen and thymus as well as the levels of plasmatic pIgR and IgA, however, the concentrations of plasmatic PGE2, Reg IV, EGFR and Akt were decreased in HT-29-bearing mice [120]. The results suggested that apoptosis of HT-29 cells induced by proteoglycan might be caused by immune-enhancement and disruption of Reg IV/EGFR/Akt signaling pathway. Jeon et al. [76] isolated an antitumor compound atractylenolide I from ethyl acetate extract of PL grown on germinated brown rice and atractylenolide I was then used to assess the inhibitory effect on HT-29 cells [76]. According to the results, atractylenolide I exhibited a strong cytotoxicity against HT-29 cells.

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Additionally, PL was also found to have a good effect on inhibiting the growth of HCT116 cells. Sohn et al. [121] found that methanol extracts of PL showed a 46.3% reduction in cell proliferation on human colorectal HCT116 cells at a concentration of 100 μg/mL.

4.1.4. Anti-sarcoma activity and anti-leukemia activity

Anti-sarcoma and anti-leukemia were another two important activities of PL based on a series of related sub-studies. Earlier in 1971, the polysaccharides from PL were reported to possess a good antitumor activity on sarcoma-180 cells [101]. Nakamura et al. [14] observed a protein-*α*-1, 3-glucan complex obtained from PL mycelia extract and firstly found that this protein-glucan complex was a potent component for antitumor activity against sarcoma-180 solid tumor cells planted in mice. Another peoteoglycan isolated from PL could inhibit the growth MCA-102 tumor and suppressed tumor weight at a dose of 100 mg/kg/d *in vivo*. It increased the ratio of $CD8^+$ DC and $CD8^-$ DC and enhanced the production of IL-12 and helper T cells (Th)-1 cytokine (IFN-*γ*) and the expression of surface molecules including MHC classes I, MHC II, CD80 as well as CD86 in MCA-102-injected mice [122]. The production of Th-2 cytokine (IL-4) by splenic cells and inguinal lymph node cells was decreased. The production IL-12 p40/p70 was significantly enhanced by peoteoglycan at 100 μg/mL and costimulatory molecule CD86 and MHC class II molecule were increased, indicating that IL-12 production in DC *in vitro* and peoteoglycan induced phenotypic maturation of DC might be mediated by TLR 2 and 4 [122]. A study was performed to compare the antitumor activity of water extracts from *Phellinus linteus*, *Phellinus baumii* and *Phellinus givus* on sarcoma-180 cells and P388 cells. The results showed all three species of PL possessed a slight antitumor effect on sarcoma-180 and P388, and *Phellinus linteus* was the most effective one against sarcoma-180 with a statistical significance [124]. Besides that, PL was shown to have an inhibitory effect on sarcoma S180 and H22, modulate immune as well as anti-oxidation [123,125].

Recent investigations indicated that PL might be a potential candidate for treating leukemia. Anti-leukemia effect of PL ethanolic extracts was evaluated on human leukemia cell line K562 *in vitro*. Experimental data

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showed B-cell lymphoma 2 and caspase 9 transcripts were remarkably increased in K562 cells. When stimulated with IFN-γ followed by treating with PL alcohol extract, transcription of the inflammatory cytokine TNF-α in K562 was elevated up by 4.5-fold, suggesting a quite strong proinflammatory effect of the PL extract on K562 cells. The high expression of TNF-α transcripts stimulated by IFN-γ treatment may also indicate a proapoptotic effect of TNF-α on K562 cells [126]. Park et al. [128] elucidated the molecular mechanisms of potent anti-leukemia activity of proteoglycan extracted from PL. The results showed the extract (0.5, 1, 1.5, 2 mg/ml) up-regulated expressions of pro-apototic Bax and Bad but down-regulated expressions of anti-apoptotic Bcl-2 and Bcl-xL. The levels of IAP family members such as XIAP and cIAP-1 were inhibited and the proteolytic activations of caspase-3 and a concomitant degradation of β-catenin were induced by proteoglycan [128]. In addition, treatment with PL significantly decreased the levels of COX-2 mRNA and protein expression without significant changes of COX-1 expression in human leukemia U937 cells [128]. PL could also have an anti-proliferative effect on NB4 human leukaemia cells. Chen et al. [127] found that a hispolon from PL inhibited cellular growth of NB4 cells in a dose-dependent manner via inducing cell cycle arrest at G0/G1 phase, which was associated with a remarkable decrease in the protein expressions of p53, cyclins D1 and E and cyclin-dependent kinases (CDKs) 2 and 4. Moreover, treatment with hispolon resulted in formation of DNA fragments and chromatin condensation and it elevatedthe levels of extrinsic apoptotic proteins (Fas and FasL), intrinsic related proteins (cytochrome c) and the ratio of Bax/Bcl-2 in NB4 cells [127]. Another study was conducted on human leukemia cell lines THP-1 and K562 cells to elucidate apoptotic induction of polysaccharides extracted from PL. Polysaccharides extract induced apoptosis of THP-1 cells in a dose-dependent way, while a pure polysaccharide had no effect like this. The polysaccharides also increased production of reactive oxygen species and potential of mitochondrial membrane, which made an important contribution to cell differentiation and death [129].

4.1.5. Other antitumor activities

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Over the past half-century, some other antitumor studies of PL were also performed in China, Japan and Korea, *etc*. Based on the researches and public data, PL was found to possess various antitumor activities besides the activities mentioned above, including lung tumor, breast tumor as well as prostate cancer (Table 2).

4.2. Antioxidant activity

A large number of studies revealed PL exhibited various bioactivities including antioxidation, immune-regulation, hypoglycemic effect, anti-inflammation, anti-obesity, *etc*. Pharmacologies of the extracts or compounds isolated from PL were summarized in Table 3. An overview of the modern pharmacological evaluations carried out on this fungus has been described in detail as follows.

A variety of chemical composition with antioxidant activity have obtained from PL. Hispidin was isolated and screened from the mycelial culture broth of PL to assesss its antioxidant effect. The results indicated that 1.0 mM hispidin exerted an antioxidant effects on super oxide anion radical, hydroxyl radical and DPPH radical at the rates of 56.8% of, 95.3% of and 85.5%, which was similar to the positive control α-tocopherol, but no significant effect on the hydrogen peroxide radical was observed [77,96]. Other investigations indicated that hispidin dimers such as hispidin with hispidin or hispolon had better effects on radical-scavenging capacity than hispidin itself [154,155]. After the fruiting bodies of PL was extracted by 70% ethanol, the obtained extract was used for antioxidative study. The ethanol extract was found to notably scavenge the stable radical DPPH with an EC_{50} of 22.07 μg/mL and inhibit lipid peroxidation with an IC_{50} of 48.45 μg/mL in a concentration-dependent manner [13]. Additionally, Park et al. [145] found water-soluble extract from cultured mycelia of *Phellinus baumii* could scavenge DPPH radical, inhibit lipid peroxidation and elevate the activities of cellular antioxidant enzymes, superoxide dismutase as well as catalase. Phelligridin G and phelligridimer A from *Phellinus igniarius* showed a good activity for inhibiting microsomal lipid peroxidation and antioxidation with an IC_{50} of 3.86 μM and 10.2 μM in rat liver, respectively [54,55]. Three phenolic compounds from the fruiting body of PL exhibited an antioxidant effect on low density lipoprotein [156]. Recently, three partially

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purified polysaccharides from PL mycelia were extracted to evaluate their antioxidant activities. They showed scavenging capacities and antioxidant activities in a concentration-dependent manner. Wang et al. [158] reported the antioxidant activity of PL was affected by the molecular weight, monosaccharide compositions and chemical structures of the polysaccharides. Assays *in vitro* showed the polysaccharides performed strong antioxidant activity including reducing power and scavenging activity against superoxide anion, hydroxyl as well as DPPH radicals [78,157].

The extract fractions using different extraction methods displayed distinct antioxidant effects. Superoxide radical scavenging activity and tyrosinase inhibitory activities of pressured extracting fractions were significantly higher than hot water and ultrasonic extracting fractions of PL [159]. The ability of chelating ferrous ions, anti-acetylcholinesterase, inhibitory activities antioxidant and xanthine oxidase of fruiting body extract was better than that from mycelia and culture filtrates [161]. Methanolic extracts from dried mycelia of *Phellinus igniarus* showed a significant antioxidant property (EC_{50} <10 mg/mL), while hot water extracting fractions exhibited reduction of power, chelation effect on ferrous ions and scavenging effect on superoxide anions with a low EC_{50} values less than 10 mg/mL [160]. The crude fractions of *Phellinus torulosus* extracted by 80% ethanol displayed a higher scavenging activity compared to quercetin and the methanol and hot water extracts performed good activites of inhibiting hydroxyl radical and hydrogen peroxide scavenging as well as reducting power on metallic compound formation [162,163].

Other researches showed extracts from PL exerted antioxidant effects through various pathways involving increasing activitiy of superoxide dismutase, simulating hepatic glutathione peroxidase, elevating generation of reactive oxygen species, *etc*., which were listed in Table 3.

4.3. Immuno-modulatory activity

PL was demonstrated to be a potent immunomodadulator. Early in 1996, Kim et al. [180] purified a polysaccharide from mycelial culture of PL and observed its immuno-regulation effect *in vitro* (100 mg/kg

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body weight) and *in vivo* (0.01 to 0.1 mg/mL). Polysaccharides from PL not only stimulated the proliferation of T lymphocyte induced by mixed lymphocyte reaction and the cytotoxicity of cytotoxic T cells responding to alloantigen but also regulated T-dependent and T-independent primary antibody responses and acted as a polyclonal activator on B cells, suggesting polysaccharide elevated the nonspecific immune functions mediated by natural killer cells and macrophages [180]. Studies demonstrated polysaccharides isolated from *Phellinus baumii* significantly enhanced proliferation of LPS-induced B lymphocyte and concanavalin A-induced T lymphocyte in a concentration-dependent manner. The polysaccharides (30 μg/mL) significantly increased the splenocyte proliferation in the absence of mitogens and markedly increased the proliferation of splenocytes pretreated with either LPS or concanavalin A at 10 μg/mL. In addition, all splenocytes treated with the polysaccharides in the presence of mitogen significantly proliferated compared to the control cells [181]. Using the optimized medium composition of exopolysaccharides from *Phellinus baumii* Pilát, the immuno-stimulating activity of exopolysaccharides was investigated. Luo et al. [179] found that the exopolysaccharides enhanced the proliferation with the highest splenocyte proliferation index (1.18±0.09, $p<0.05$) at 100 μg/mL in the absence of mitogen and increased the proliferation index (1.19±0.04, $p<0.01$) to control at the same concentration in the presence of LPS [179]. A water-soluble polysaccharide isolated from *Phellinus baumii* increased the proliferation of RAW264.7 cells in a dose-dependent manner. The proliferation rate even reached at 198.1% when the concentration was elevated up to 400 μg/mL [105]. To compare the immunomodulating properties among aqueous, ethanol, and polysaccharides extract from the cultured PL fruiting bodies on LPS-induced human monocytic (THP-1) cells, Wu et al. [182] performed a study and elucidated polysaccharides were the most active compositins with the reason that polysaccharides could remarkably block myeloid differentiation factor 88 (MyD88), NF-κB expression and cytokine production in LPS-stimulated cells and suppressed the activation and phosphorylation of c-Jun N-terminal kinase and p38. The results suggested that PL could influence MyD88-dependent and mitogen-activated protein kinase signaling pathways.

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Additionally, the stimulation of a novel polysaccharide-protein complex extracted from PL (100 μg/mL) increased the B cell proliferation for 8-fold higher than that of the unstimulated, but T cells were not affected. Another study found that the extract and LPS acted on B cells in a similar mode, but the extract-induced cellular activation was not inhibited by polymyxin B (a specific inhibitor of LPS) [147]. Polysaccharides from PL fruiting bodies (32.6 pg/mL) showed an immunosuppressive effect on activated human peripheral blood mononuclear cells (PBMCs) and synthesis of IFN-γ [178]. Reports on the immuno-modulatory activity of PL are abundant and most part of them have been described in Table 3.

4.4. Hypoglycemic Activity

Polysaccharides contribute significantly to hypoglycemic activities of PL. Polysaccharides purified from mycelial culture of PL inhibited the development of autoimmune diabetes in female non-diabetic mice. Treatment with polysaccharide at 30 mg/kg on female non-diabetic mice every other day from 8 to 24 weeks age prevented the onset and progression of autoimmune diabetes and infiltration of immune cells into the islets and the expression of inflammatory cytokines was also inhibited. The mean levels of blood glucose in drug treated group was reduced to 110 mg/dL, but non-diabetic control group was 499 mg/dL and histological observation of the pancreatic islets, revealing that islets obtained from PL-treated mice were less infiltrated with lymphocytes compared with control group [204]. PL apparently suppressed autoimmune responses to β-cells in pancreatic islets through blocking Th1 cytokine expression in non-diabetic mice. Nevertheless, STZ-induced diabetic development couldn't be prevented by the PL-treatment ICR mice [204]. Proteomic and transcriptomic analysis on the STZ-induced diabetic Sprague-Dawley rats demonstrated that carbonyl reductase, hydroxymethylglutaryl-CoA synthase and putative human mitogen-activated protein kinase activator with WD repeats-binding protein in diabetic pancreas were down-regulated significantly. However, four proteins in the diabetic rats were reverted to normal levels by the treatment with PL extracellular polysaccharides at 200 mg/kg body weight daily for 14 days [196]. The transcript levels of many genes were also positively altered and

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it played a preventive role in *β*-cell destruction mediated by STZ. According to an earlier study, extracellular polysaccharides from PL decreased plasma glucose in STZ-induced SD rats by 49% at a dose of 100 mg/kg body weight and reduced the concentrations of total cholesterol, triacylglycerol and activity of aspartate aminotransferase by 32%, 28% and 20%, respectively [197]. Oral administration mycelia polysaccharides of PL (100 mg/kg/d) remarkably reduced blood glucose by 35.6% in alloxan-induced diabetic mice [198]. By experimented on male C57BL/6J *ob/ob* mice, exopolysaccharides from PL (200 mg/kg body weight) descended plasma glucose levels by 32% and notably activated peroxisome proliferator-activated receptor gamma (PPAR-γ) [202]. From the results, the expolysaccharides improved insulin sensitivity by regulating PPAR-*γ*-mediated lipid metabolism to exert ahypoglycemic effect. Water extract of PL were orally administrated to the pancreatectomized SD rats (500mg/kg) and the outcome indicated a lower fasting serum glucose levels and amelioration of insulin resistance [201].

Hispidin in PL was also illustrated to exhibit antidiabetic effect through preventing *β*-cells from the toxic action of reactive oxygen species in diabetes, which showed a similarity in inhibiting hydrogen peroxide-induced apoptosis and increase insulin secretion in hydrogen peroxide-treated cells [168]. Decreased levels of plasma glucose and blood serum insulin as well as inhibited a-glucosidase activity of diabetes by PL were found as well [199,200,203,205].

4.5. Anti-inflammatory activity

Although various activities of polysaccharides from PL were found, no evidence was found to disclose the relationship between anti-inflammatory activity and the polysaccharides. The *n*-BuOH extract was demonstrated to perform an excellent anti-inflammatory activity in croton oil-induced ear edema test, anti-nociceptive activity and good inhibitory effect on the chick embryo chorioallantoic membrane (CAM) angiogenesis. More improtantly, *n*-BuOH extract showed almost equivalent anti-inflammatory effect to indomethacin. Oral administration of *n*-BuOH extract of PL (100 mg/kg) could inhibit the number of writhing

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by 35.9% compared to indomethacin 70.4% [213]. Additionally, *n*-BuOH subfraction of the ethanol extract prepared from PL fruiting bodies showed an anti-inflammatory effect by inducing heme oxygenase-1 (HO-1) of the RAW264.7 macrophages. Treatment with *n*-BuOH subfraction at 0.1, 0.25 and 0.5 mg/mL elevated HO-1 expression in non-stimulated RAW264.7 macrophages in a concentration-dependent manner. Pretreatment with subfraction concentration-dependently (0.01, 0.1, 0.5 and 1 mg/ml) diminished LPS-stimulated iNOS promoter activity and notably reduced iNOS expression and NO production in macrophages. The subfraction decreased the c-Jun N-terminal kinases (JNKs) activation stimulated by LPS in the macrophages [207]. Furthermore, mechanisms of these effects were revealed by the subsequent researches. The results indicated thatinduction of HO-1 expression was reduced by GF109203X (a general inhibitor of PKC) and rottlerin (a specific PKCδ inhibitor), but not by Gö6976 (a selective inhibitor for PKCα/β) [206]. The phosphorylation of protein kinase Cδ (PKCδ) but not PKCα/β II was enhanced in a time-dependent manner by subfraction. Additionally, PL suppressed the expression of matrix metalloproteinase-9 (MMP-9) and its promoter activity, which depended on PKCδ [206]. The findings suggested that PL up-regulated HO-1 via PKCδ/Nrf2/ARE signaling way to exert anti-inflammatory effect in RAW264.7 macrophage cells. Another possible mechanism of inhibiting the NO and PGE2 production was down-regulating iNOS and COX-2 gene expression via ROS-based NF-κB and MAPKs activation. PL *n*-BuOH subfraction obviously inhibited nuclear translocation of NF-κB p65 subunits as well as IκBα phosphorylation and degradation. It also suppressed the activation of mitogen-activated protein (MAP) kinases including p38 and stress-activated protein kinase/c-Jun NH2-terminal kinase (SAPK/JNK). In addition, PL (25-100 μg/mL) inhibited iNOS mRNA expression before exposure to 1 μg/mL LPS and intracellular accumulation of reactive oxygen species was directly blocked in RAW 264.7 cells at a concentration of 50 μg/mL. COX-2 expression was restrained to inhibit PGE2 production at the transcriptional step in LPS-activated RAW 264.7 macrophages. Huang et al. [82] validated inotilone isolated from PL (0, 6.25, 12.5, and 25 μM) showed a significant inhibition of NO production and iNOS protein expressions

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concentration-dependently. They also found that expressions of iNOS, NF-*κ*B and MMP-9 as well as ERK, JNK and p38 phosphorylation in LPS-stimulated RAW264.7 macrophages were inhibited by inotilone. Inotilone (5 mg/kg) significantly decreased Carr-induced iNOS, cyclooxygenase-2 (COX-2), NF-*κ*B, TNF-*α* level and MMP-9 expression at the 5th hour in the edema paw and reduced the paw edema induced by Carr administration (10 mg/kg) at the 4th and the 5th hour after treatment, but increased the activities of catalase (CAT), superoxide dismutase (SOD), and glutathione peroxidase (GPx) [82]. However, intraperitoneal injection inotilone could result in decrease of neutrophil infiltration into sites of inflammation. Several phenolic compounds were isolated from *Phellinus baumii* and their anti-inflammatory activities were assessed. It was found that pretreatment with phelligrin A, methylphelligrins A and B substantially inhibited LPS-induced NF-*κ*B activation in human prostate cancer cells with IC_{50} values of 54.50 μM, 36.44 μM, and 22.46 μM, respectively [83]. Pretreating intraperitoneally with extracts of *Phellinus gilvus* and *Phellinus baumii* at 20 mg/kg body weight could reverse the total hemamebas, neutrophils and the level of IL-1*β* in LPS-stimulated rats, which was similar to those in saline-challenged rats [208]. Post-administration of proteoglycan derived from PL (200 mg/kg) lessened anti-CII IgG and IgG2a antibodies (Abs) as well as various kinds of cytokines including IL-12, TNF-*α*, and IFN-*γ*, which resulted in inhibition of collagen-induced arthritis in mice [209]. Treatment with the aqueous extracts of *Phellinus baumii* (10 μg/mL) remarkably decreased expressions of endoplasmic reticulum stress markers, iNOS and NO in LPS-treated RAW264.7 [211]. PL grown on germinated brown rice (200, 500 mg/kg/day) notably reduced the production of NO and prostaglandin E2 (PGE2) concentration-dependently in RAW 264.7 macrophages, and ameliorated the pathological characteristics of colitis in dextran sodium sulfate-induced mouse model [210]. PL was also found to perform anti-inflammatory by regulating the expressions of iNOS, COX-2, YNF-*α* and their mRNAs as well as inflammation-related protein expressions such as NF-*κ*B p65 and inhibitory-*κ*B*α* in both HepG2 cells and rats, [212,214].

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4.6. Other bioactivities

According to clinical practice of TCM, PL were widely used for treating blood diseases in Chinese ancient times. In recent decades, PL has been found to be effective on a diversity of diseases besides the activities described above. Some relevant investigations were carried out using the crude extracts or purified compounds from PL (Table 3).

5. Conclusion

PL is a well-known medicinal fungus and has been extensively used in China, Japan and Korea, *etc*. It has been conventionally used to improve circulation of blood, hemostasis and the diseases related female menstruation for several centuries. Modern pharmacological studies demonstrated PL was a potential candidate drug for antitumor. A diversity of bioactivities of PL *in vitro* and *in vivo* were found, including anti-cancer, immuno-regulation, anti-diabetes, anti-oxidation, anti-inflammation, *etc*. Based on reported materials, polysaccharides were the main chemical composition of PL and they were responsible for various pharmacological effects of PL. Considering its excellent antitumor activity, various bioactivities and resource scarcity, further fermentation studies is indispensable to improve the production and extend its applications. The optimized fermentation conditions were recommended as adequate carbon sources and nitrogen sources, pH 5-6.5, relative humidity excess of 96%, fermented temperature about 30 °C, *etc*.

In recent years, numerouspolysaccharides and proteoglycan with excellent bioactivities and antitumor effects have been isolated and obtained from medicinal fungi. As a medicinal fungus, PL has much scientific value and clinical potential for developing new drugs, especially antitumor drugs. In the past decades, abundant investigations on fermentation, phytochemistry and pharmacology of the PL were performed. However, the following proposed points should be still highlighted for future studies. First and importantly, to ensure clinical medication safety, the toxicity of PL both *in vitro* and *in vivo* is urgent to reveal. Second, the taxonomy and names of species and genus of PL are still not clear and unified, which is required to clarify and elucidate in

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detail. Third, traditional uses based on empirical practical descriptions of PL was mainly applied for treating diseases related on blood circulation and female menstruation in the ancient days, but pharmacological studies showed it made preeminent performances on antitumor, immuno-regulation, antioxidant, hpyerglycemic and anti-inflammatory effects. Therefore, further stidies is necessary to elucidate the relationship between traditional uses and modern pharmacological activities. More importantly, the relationships between antitumor effects and other pharmacological effects should be illustrated. Last, althrough modern pharmacological studies of PL *in vivo* and *in vitro* were thorough and abundant, the researches on the clinical explorations were inadequate. Hence, numerous clinical studies should be further conducted.

This review provided comprehensive and detailed information of PL and systematically presented its traditional uses, fermentation, phytochemistry and pharmacology. Critical analysis and scientific gaps were also analyzed in the text. As is mentioned above, further investigations need to be performed based on the published reports and review. Fortunately, the current review can provide a reference for peer researchers and stimulate the exploitation and utilization of PL.

Conflict of interest

The authors declare that they have no conflict of interest.

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Figure 1

Fig. 1. The plant and medicinal materials (fungus parasitizing on the trees (a) the fruiting body (b) and decoction pieces (c)) of PL.

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Figure 2

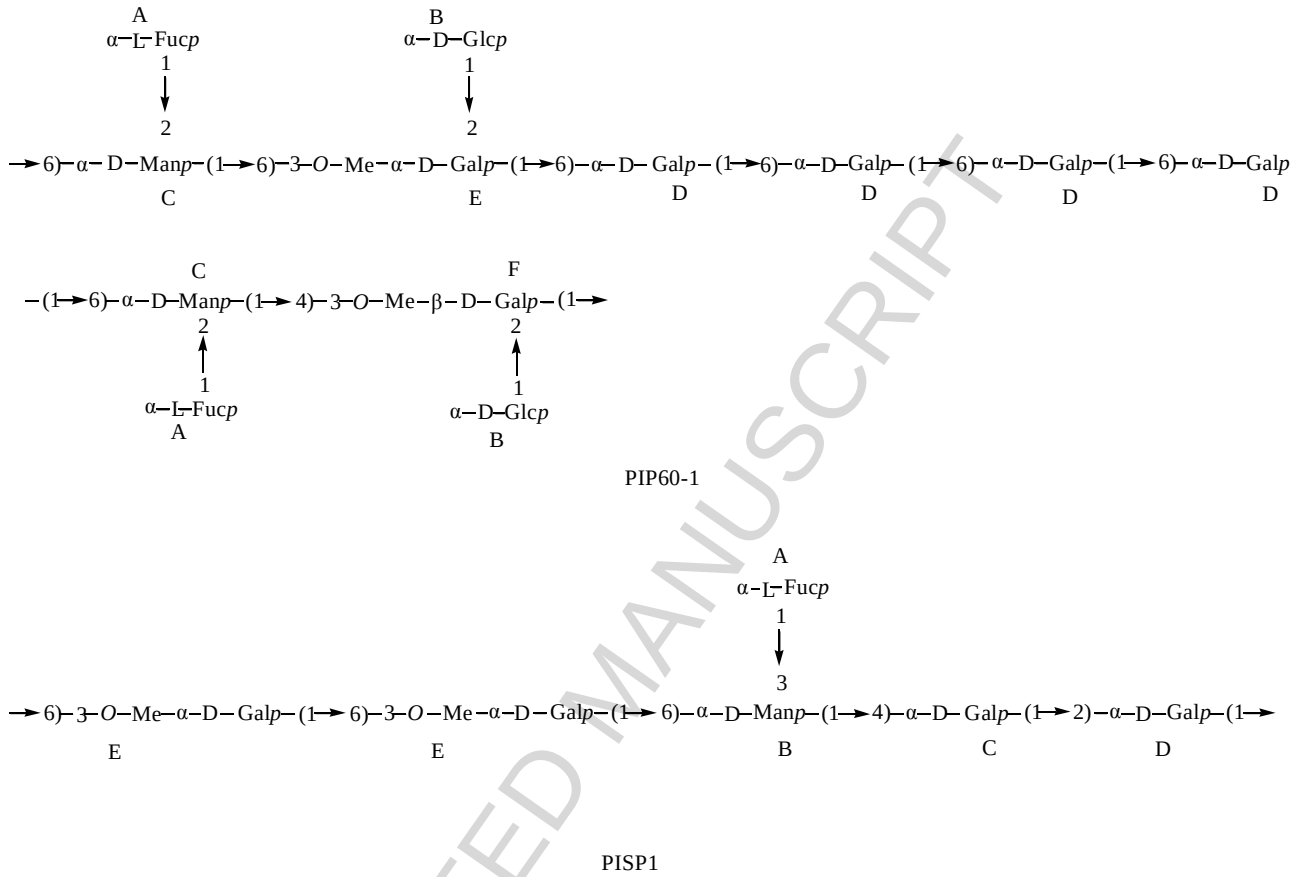

Fig. 2. The chemical structure of PIP60-1 and PISP1 from *Phellinus igniarius*.

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Figure 3

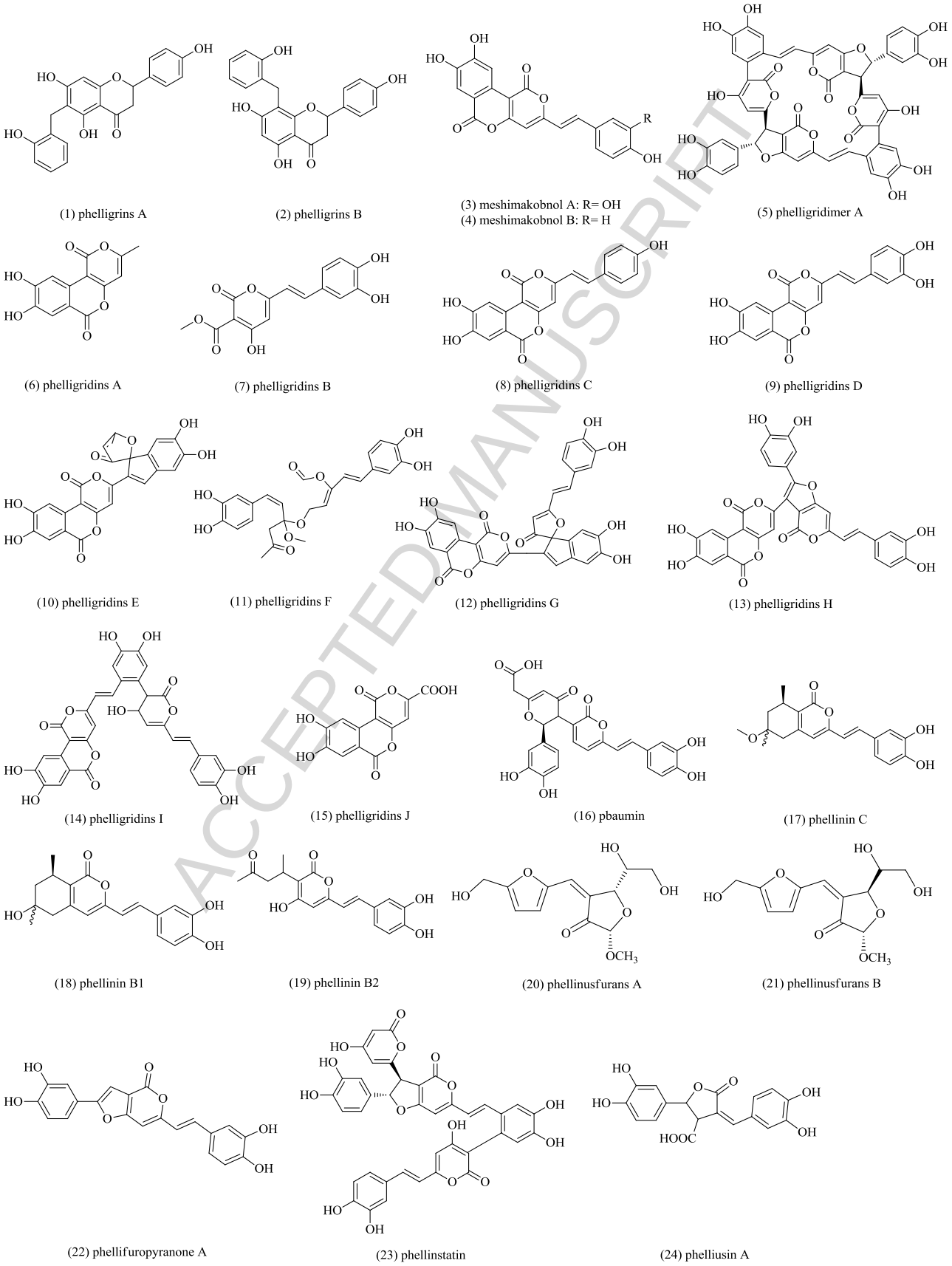

Fig. 3. The chemical structure of flavones, pyranones and furan from PL.

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Figure 4

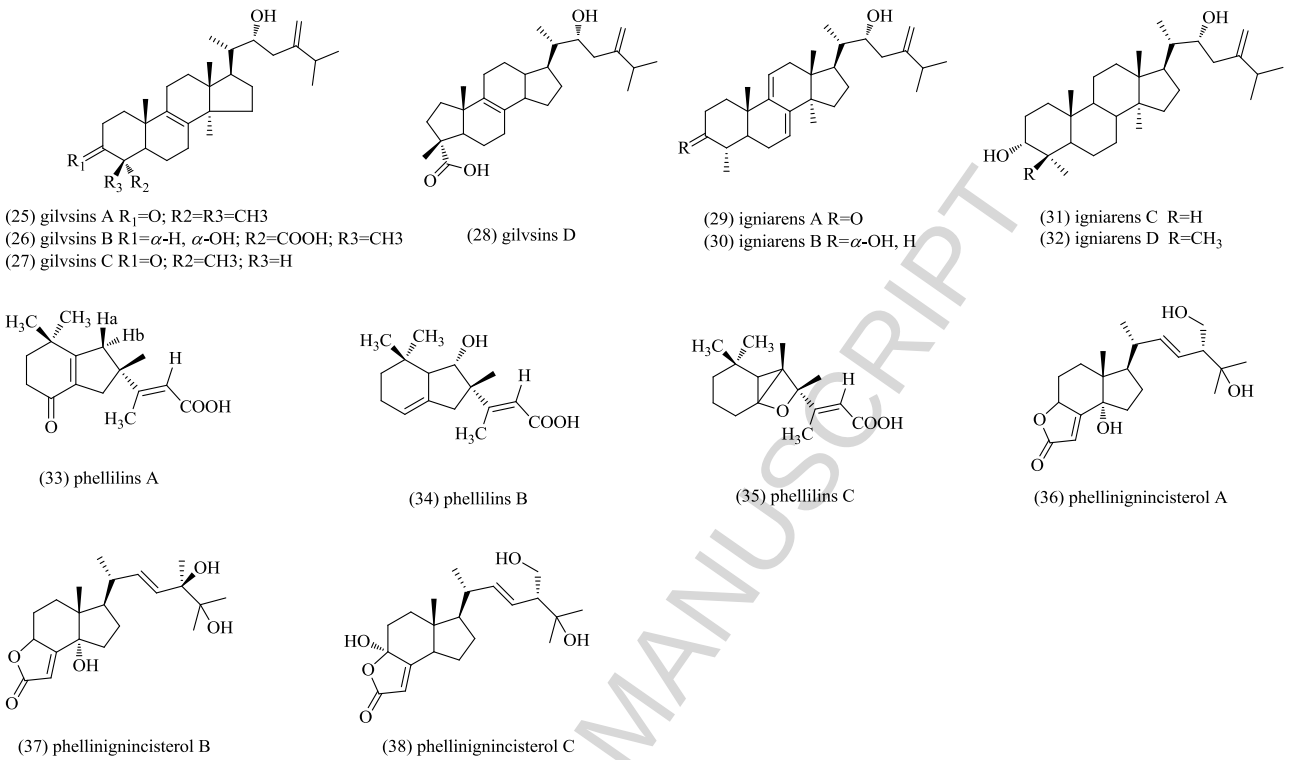

Fig. 4. The chemical structure of terpenes and steroids from PL.

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Figure 5

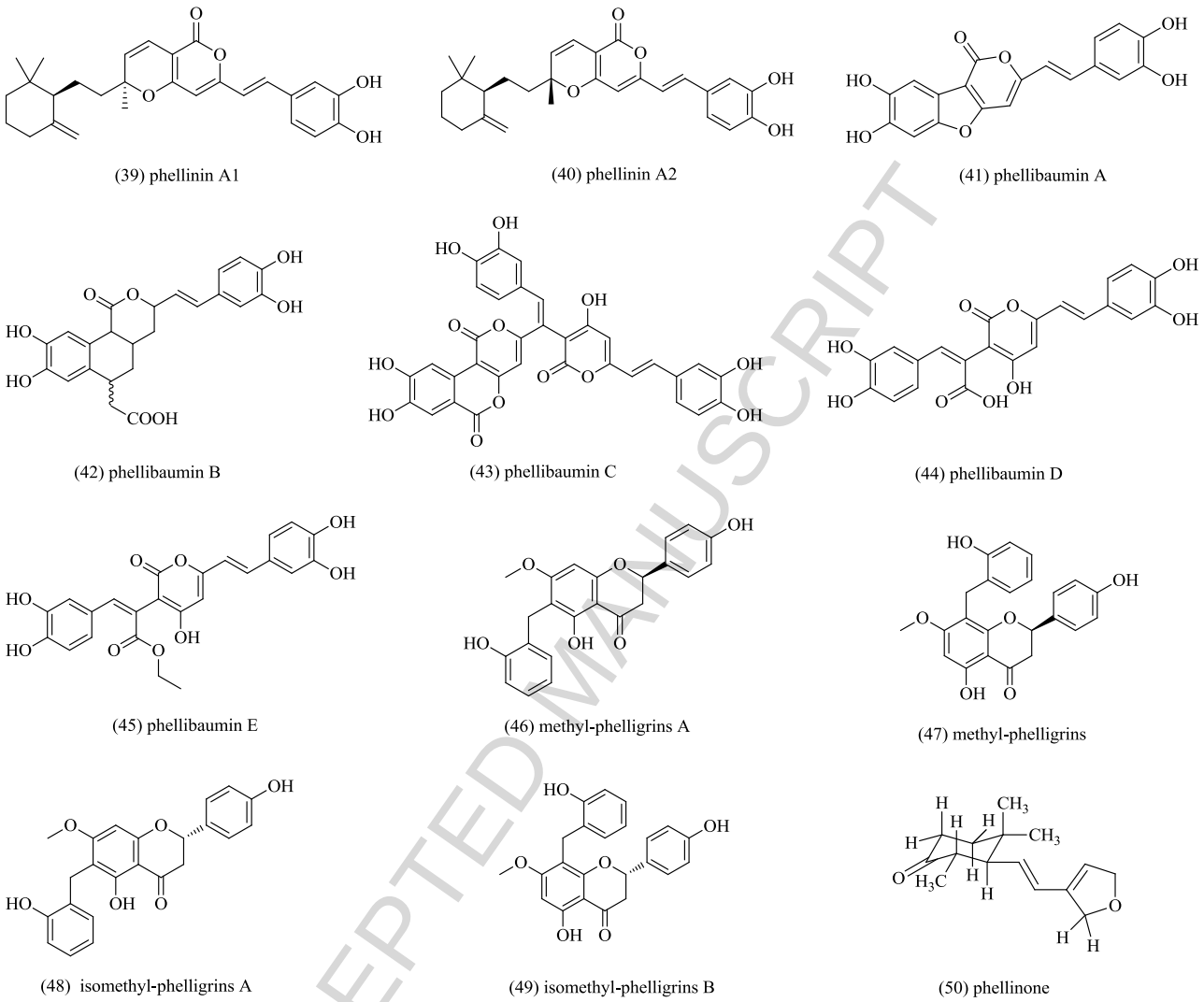

Fig. 5. The chemical structure of other compounds in PL.

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Table 1

Chemical compounds have been isolated from *Phellinus linteus.*

No.	Chemical name	Origin	Reference(s)
1	Acid proteoglycan	The fruiting body of PL (Berk. & M.A. Curtis) Teng	[32]
2	Exopolysaccharide	Submerged culture of PL	[34]
3	Fucoglucan PBF4	The fruiting body of *P. baumii* Pilat	[35]
4	Polysaccharide PLP	Mycelial culture of PL	[36]
5	Polysaccharide	The fruiting body of PL	[37]
6	β-D-glucan PBF1	The fruiting body of *P. baumii* Pilát	[38]
7	Polysaccharides	The fruiting body of PL	[39]
8	Heteroglucan	The fruiting body and cultured mycelia of PL	[40]
9	A water-soluble glycan	Water extract of PL	[6]
10	Heteropolysaccharide named PISP1	The fruiting body of *P. igniarius*	[41]
11	Polysaccharides contained (1-3)-β-glucans with a (1-6)-linkage	The fruiting body of PL	[42]
12	Heteropolysaccharide PIP60-1	The fruiting body of *P. igniarius*	[43]
13	A water-soluble heteropolysaccharide termed PBF2	The fruiting body of *P. baumii* Pilát	[44]
14	Polysaccharide Piw-1 containing 3-O-methyl galactose	The fruiting body of *P. igniarius*	[45]
15	12-hydroxy-7-oxo-5,8,11,13-teraene-18,6-abietanolide, Stigmasta-7,22-diene-3β,5α,6α-triol	The fruiting body of *P. igniarius*	[46]
16	A homopregnene, four steroid, three heptanorergosterane derivatives and nine tremulane sesquiterpenes	Cultures of *P. igniarius*	[47]
17	Meshimakobnol A and B	The fruiting body of PL	[48]
18	β-(1→3)-D-glucan	The fruiting body of PL	[49]
19	Phelliribsin A	The fruiting body of *P. ribis*	[50]
20	Phellifuropyranone A, phelligridin G	The fruiting body of PL	[51]
21	Hispolon and phelligridins C,D,E,F, (E)-4-(3,4-dihydroxyphenyl)but-3-en-2-one, 4-hydroxybenzaldehyde, protocatechualdehyde, syringic acid, protocatechuic acid, caffeic acid, isoergosterone, octadecyl ferulate	The fruiting body of *P. igniarius*	[52]
22	Phellinins A1 and A2, hispidin and 1,1-distyrylpyrylethan	Cultured broth of *P.* sp. KACC93057P	[53]
23	Phellinone	Cultured broth of PL	[54]
24	Sphingolipid, protocatechualdehyde, 5-hydroxymethyl-2-furaldehyde, succinic acid, and fumaric acid	The fruiting body of PL	[55]
25	Phelligridins H, I, and J, davallialactone, scopolin, nebularine, uridine, trehalose, glucitol, and ethyl glucoside	The fruiting body of *P. igniarius*	[56]
26	Benzyl dihydroflavones phelligrins A and B	The fruiting body of *P. igniarius*	[57]

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27	Sesquiterpenes, phellilins A–C	Mycelium of PL	[58]
28	Two indole derivatives 7-methoxyindole-3-carboxylic acid methyl ester and 1-methylindole-3-carboxaldehyde, and a phenolic (E)-4-(3,4-dihydroxyphenyl)but-3-en-2-one	The fruiting body of PL	[59]
29	Phellinusfurans A and B	The fruiting body of PL	[60]
30	Polysaccharide	The fruiting body of *P. baumii* Pilát	[61]
31	Polysaccharide R1	The fruiting body of PL	[62]
32	Two neutral heteropolysaccharides	The fruiting body of PL	[63]
33	5 flavones (Naringenin, sakuranetin, two hydrogen kaempferide, 7-methoxy two hydrogen kaempferide, eriodictyol) and 2 coumarin coumarin and scopoletin	The fruiting body of *P. igniarius*	[64,65]
34	Phelligrins A and B	The fruiting body of *P. igniarius*	[66]
35	Meshimakobnol A and B	The fruiting body of PL	[48]
36	Baumin, davallialactone, hispidin, hypholomine, interfungin A, inoscavin A, and phelligridin D	The fruiting body of *P. baumii* Pilát	[67]
37	Gilvsins A, B, C, D, and 24-methylenelanost-8-ene-3β, 22-diol and 5α-ergosta-7,22-diene-3-one	The fruiting body of *P. gilvus*	[68]
38	Igniarens A, B, C and D	The fruiting body of *P. igniarius*	[69]
39	Eugenol, protocatechualdehyde, protocatechuic acid, caffeic acid, iso-ergosteion, cluytyl ferulate, microelements	The fruiting body of *P. igniarius*	[70]
40	Heteropolysaccharide-proteins	Culture broth of PL	[71,72]
41	β-pyrano polysaccharide termed HLM	Culture broth of *P. igniarius*	[73]
42	Three sesquiterpenes, three steriods, nine other compounds	Culture broth of *P. igniarius*	[74]
43	Hispidin	Culture broth of PL	[75]
44	Atractylenolide I	PL grown on germinated brown rice	[76]
45	Hispidin and its dimers	Culture broth of PL	[77]
46	Polysaccharides	mycelial culture of *P. igniarius*	[78]
47	Phelligridimer A	The fruiting body of *P. igniarius*	[79]
48	Phelligridin G	The fruiting body of *P. igniarius*	[80]
49	Phelligridin D, 5-dihydroergosterol,1,2-benzenedicarboxylic acid bis (2-methyl heptyl) ester, protocatechuic acid, cirsiumaldehyde, uracil, gallic acid, 2,5-dihydroxybenzoic acid, ferulic acid, 2,3-dihydroxybenzaldehyde, arbutin, isoferulic acid, ellagic acid	The fruiting body of PL	[81]
50	Inotilone	The fruiting body of PL	[82]
51	Phellibaumins A-E, methylphelligrin A, *epi*-methylphelligrin A, methylphelligrin B, *epi*-methylphelligrin B	The fruiting body of *P. baumii* Pilát	[83]
52	Davallialactone, methyldavallialactone, hypholomine B, interfungins A, interfungins A, and inoscavin A	The fruiting body of PL	[26,84,85]
53	Hypholomine B	Culture broth of PL	[86]

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54	4-(3,4-dihydroxyphenyl)-3-buten-2-one	Fermentation broth of PL	[87]
55	Phellinstatin	Culture broth of PL	[88]
56	Inoscavin E and C	The fruiting body of PL	[89]
57	Phellibaumin A, B, C, D and E, methylphelligrins A and B, isomethylphelligrins A and B, (22E,24R)-6β-oxethyl-ergosta-7,22-diene-3β,5α-diol and (22E,24S)-6β-oxethyl-ergosta-7-ene-3β,5α-diol	The fruiting body of *P. baumii*	[83]
58	Methyl-4-hydroxyphenylacetate, salicylaldehyde, 2,5-bis(hydroxymethyl)furan, succinic acid, 2-methylol-5-methoxylfuran, and N-acetyl-2-p-hydroxyphenylethylamine	The mycelial culture broth of PL	[90]
59	Phelligridins A1 and B2	The fruiting body of *P. igniarius*	[91]
60	Phellinin B and phellinin C	The mycelial culture broth of PL	[92]
61	Phellinsin A	Culture broth of *P.* Sp. PL3	[93]

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Table 2

The anti-tumor activity of extracts or compounds from PL.

No.	Anti-tumor activities	Extract/ compound	Test living system	Routes of administration/dose	Reference(s)
1	Anti-hepatocarcinoma activity	Ethanolic extract	Human hepatocarcinoma SK-Hep-1 cells and rat heart vascular endothelial cells	25-100 μg/ml	[102]
2		Water extract	Hep3B cells	100, 300, 500 μg/ml	[103]
3		Hispolon	Hep3B cells, J5, HepG2	45 μM	[104]
4		A water-soluble polysaccharide	Human liver cancer HepG2 cell lines	0, 100, 200, 400 μg/mL	[105]
5		Mycelial culture	SCID CB-17 mice received a transplant of Hep3B cells	i.g. 50, 100, 250 mg/kg/d	[106]
6		Proteoglycan	HepG2 cells	200 μg/ml, and p.o. 200 mg/kg	[107]
7		Water extract	Hep-2 and SK-MS-cells	5 and 10 mg/ml	[108]
8		Phellinus linteus	HepG2 cells	0–1.0 mg/mL	[109]
9		Endo-polysaccharide	S180 and H22 cell lines	i.g. 500, 250, 125 mg/kg	[110]
10	Anti-melanoma activitiy	Extract of Phellinus linteus	C57BL/6 mice implanting H22 and B16 melanoma cells	i.g. 40, 200, 1000 g/kg/d	[111]
11		Methanol extract	B16-F10 melanoma cells	10, 50, 100 μg/ml	[112]
12		Polysaccharides	B16F10-implanted mice; NCI-H23-implanted nude mice; B16F10 and NCI-H23 cells cells	i.g. 100 mg/kg; 0.3 to 30 μg/ml	[113]
13		Ethyl acetate extract	B16F10 cells	0, 1, 10, 100, 500 μg/ml	[114]
14		Polysaccharides	HT1080 cell line	Various concentrations	[42]
15	Anti-melanoma and anti-metastatic activities	Aqueous extract	B16BL6 cells, C57BL/6 mice injiected with B16BL6 cells	125, 250, 500, 1000 μg/ml i.g. 2, 20, and 50 mg/kg	[115]
16	Anti-metastatic, anti-invasion and cytotoxic effects	Acidic polysaccharide	B16F10 murine melanoma cells and Female C57BL/6 mice	i.p. 10, 30, and 100 mg/kg; 10, 30, or 100 μg/m;0.03 to 30 μg/ml	[116]

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17	Anti-colorectal carcinoma activity	Polysaccharides	SW480 colon cancer cells, HT1080 fibro-sarcoma cells	125, 250, 500, 1000 μg/mL	[117]
18		Protein-bound polysaccharide	SW480 human colon cancer cells	125, 250, 500 μg/mL	[118]
19		Polysaccharide	HT-29 (colorectal carcinoma) cells, Caco-2 (colorectal carcinoma) cells, LO-2 (human hepatocyte) cells; Female BALB/c-nu/nu mice	25, 50, 100, 200 g/ml; i.g. 100, 200 mg/kg	[119]
20		Proteoglycan	HepG2, HT-29, NCI-H460 (lung cancer cell), MCF7 (breast cancer cell), L-929 (mouse fibroblast cell) cells , and Female BALB/c-nu/nu mice	15.625, 31.25, 62.5, 125, and 250 μg/ml; i.g. 50 and 100 mg/kg	[120]
21		Atractylenolide I	HT-29, CCD-986sk cell lines	IC_{50}=69.8 μg/ml	[76]
22		Methanol extract	Human colorectal HCT116 cell, human embryonic kidney cell HEK-293, pre-adipocyte cell 3T3-L1, and pre-osteoblast cell MC3T3-E	100 μg/ml	[121]
23	Anti-sarcoma activity	Proteoglycan	Female C57BL/6 mice; MCA-102 tumor cells	i.g. 100 mg/kg/day and 100μg/ml	[122]
24		Hot water and alkali water extract	Male ICR mice and SD rats implanted with Sarcoma 180 cells	20 mg/kg/d	[123]
25		Water extract	Sarcoma 180 and P388 cells	7.5, 15, and 30 μg/ml	[124]
26		Crude polysaccharides	Swiss Albino mice implanted with Sarcoma 180 cells	i.g. 50, 115, 150, 200 mg/kg	[101]
27		Extract of mycelia	Female specific pathogen-free BDF-1 mice implanted with Sarcoma-180 solid tumor cells	Unknown	[14]
28		Crude extracellular polysaccharides	Mice implanted with sarcoma S180 and H22 cancer cells	i.g. 100, 200, 400 mg/kg	[125]
29	Anti-leukemia activity	Ethanol extract	Human erythroid leukemia cell line K562	1, 5, and 10 μg/mL	[126]
30		Hispolon	Human hepatocellular carcinoma NB4 cells	0.625, 1.25, 2.5, 5, and 10 μg/mL	[127]
31		Proteoglycan	Human leukemia U937 cells	0.5, 1, 1.5, 2 mg/ml	[128]
32		Polysaccharide	Human leukemia cell lines THP-1 and K562	2-1000 μg/mL	[129]

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33	Anti-pulmonary carcinoma activity	Ginseng extract fermented with *Phellinus linteus*	A549, L1210, NIH3T3, and Raw 264.7 cells; C57BL/6 implanted eith B16F10 and ICR mice	0-200 µg/mL	[130]
34		Aqueous extract	Human lung carcinoma A549 cells	0-5 mg/ml	[131,132]
35		Mycelial extract	Human lung carcinoma A549 cells	0-5 mg/ml	[133]
36	Anti-prostate cancer activity	*Phellinus Linteus* powder	Human prostate cancer PC3 cells, DU145 cells; nude mice with DU145 or PC3 cells	1 mg/ml or 2 mg/ml; i.g. 30 mg/kg	[134]
37		*Phellinus Linteus* powder	Human prostate epithelial PrEC cells and prostate cancer LNCaP and PC3 cells	0.75 and 1 mg/ml	[135]
38		*Phellinus Linteus* powder	Human LNCaP and PrEC cells	0.25, 0.5, 0.75, 1 mg/ml	[136]
39	Anti-breast cancer activity	Aqueus extract	Human breast cancer cells (MCF-7, MDA-MB-231) and human prostate cancer cells (PC-3, LNCaP)	0-1.0 mg/ml	[137]
40		Hispolon	MCF7, T24, J82 cells, and MDA-MB-231 cells	10, 20, 30, 40, 50, 60 µg/mL	[138]
41		Hispolon	Human epidermoid KB cells	0-16 µg/mL	[139]
42	Anti-proliferative activity	A novel extracellular laccase	Numan HepG2 and mouse lymphocytic leukemia L1210	0-10 µM	[140]
43	Anti-tumor effect	Phellinus linteus extract	ICR mouse injected with S-180 cells	i.v. 31.3-250 mg/kg	[141]
44		Acidic polysaccharide	C57BL/6 mice implanted with B16 melanoma cells	10, 50 and 200 µg/ml	[142]
45		Hot water extract of fruiting body	Male BALB/c mice implanted with CT-26 cell line	250, 500, and 1,000 mg/kg	[143]
46		Crude polysaccharides	EC109 and SiHa tumor cells	350, 750, 1500 µg/mL	[144]
47		Phelligridin G	A 2780 and HCT-8 cells	IC_{50}= 20.4 and 30.2 µM	[80]
48		Water Extract of culture borth	Chinese hamster lung fibroblast, V79-4 cells	0-100 µg/mL	[145]
49		Methanol Extract	B16/F10 melanoma and NIH3T3 cells	0.625, 12.5, 25, 50, 100 µg/mL	[146]
50		Polysaccharides-protein complex	B16 melanoma cells	100 µg/mL	[147]

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51	Mycelium Extract	L-929 cells	0.001. 0.01, 0.1 μg/mL	[148]
52	Ethanol extract	HepG2 and A549 cells	Content: 0.5%-4.0%	[149]
53	The fruiting body	B16/BL6-bearing male C57BL/6J mice	i.g. 83, 415, 2075 mg/kg/d	[150]

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Table 3

The bioactivities of extracts and compounds from *Phellinus linteus*.

No.	Pharmacological activities	Extract/ compound	Test living system/ Method	Routes of administration/ dose	Reference(s)
1	Anti-oxidant activity	Hispidin	H_2O_2, DPPH radical, and superoxide anion scavenging activity	0.01, 0.1, 1 mM	[154]
2		Hispidin dimers	DPPH assay, and superoxide anion scavenging activity	10, 25, 50, 100 µg/ml	[155]
3		EtOAC and *n*-BuOH extracts	LDL-axidantion activity	0-6 µg/ml	[156]
4		Polysaccharides	DPPH, TEAC assay, ferric reducing ability of plasma assay	0-2.0 mg/mL; 0.1 mL; 10, 20, 40, 80 µg/mL	[157]
5		Polysaccharides	Superoxide anion, DPPH, H2O2 scavenging activity, and reducing power activity	0-3500 mg/L	[158]
6		hot water extraction, ultrasonic extraction, and pressured extraction	ORAC; DPPH radical, ABTS radical, Superoxide radical, FRAP	100, 250, 500, 1000 µg/ml	[159]
7		Hispidin and its dimers	DPPH radical, ABTS radical, Superoxide radical scavening assay	0.8, 4, 20, 100 µmol/L	[77]
8		Methanolic extractions of mycelia	Reducing power; DPPH and superoxide anion scavenging activity; ferrous ions Chelating ability	0-20 mg/mL	[160]
9		Total phenols and flavonoid	DPPH assay, ferrous ions Chelating ability, XO inhibitory effect	25, 50, 100, 200, 400 µg/ml	[161]
10		Phelligridimer A	Rat liver microsomal lipid peroxidation inhibition effect	IC50=10.2 µM	[79]
11		Phelligridin G	Rat liver microsomal lipid peroxidation inhibitiion effect	IC50=38.6 µM	[80]

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12	Methanol and hot water extraction	DPPH radical, Non-site-specific hydroxyl radical, Site-specific hydroxyl radical, and H2O2 radical scavenging activity, reducing power activity	50, 500, 1000 μg/ml	[162]
13	Hispidin	Superoxide radical, DPPH radical, and ABTS radical scavening assay	Mean IC50=2.5~14.1 μM	[96]
14	Crude extracts and crude fractions	DPPH radical and and total phenolic content scavening assay	Mean IC50=24.1~125.9 μg/ml; 15.0~32.7 mg	[163]
15	Polysaccharides	TEAC and FRAP assay; Hydroxyl radical scavening assay	0.05–1.5 mg/mL	[78]
16	70% ethanol extraction	DPPH assay, LPO assay, XO activity inhibition assay	1, 10, 100, 1000 μg/ml; 10-300 μg/ml; IC50=146.80	[13]
17	Water-soluble extraction of mycelia	Reactive oxygen species and DPPH scavening, LPO assay, SOD activity	10, 50, 100 μg/ml	[145]
18	Ethanolic extraction	DPPH, XO, SOD, ABTS inhibitory activity	0–1000 μg/mL	[164]
19	Ethanolic and hot water extractions	DPPH assay; ferrous ions chelating effect; Reducing power	0.5–20 mg/ml; 0.01–20 mg/ml; 0.5–20 mg/ml	[165]
20	Extract of Phellinus linteus	DNA damage induced by ROS	50 and 100 μg/ml	[166]
21	Water extraction	SOD and TBARS assay, effect on hepatic glutathione peroxidase; Male SD rats	i.g. 50 and 100 mg/kg	[167]
22	Hispidin	DPPH, superoxide, H2O2 radical, and ROS scavenging activity	5, 10, 20, 30, 40 μM	[168]

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23	Hispidin	peroxynitrite-mediated DNA damage and hydroxyl radical generation	1-20 μg/ml	[169]
24	Methanol Extraction	DPPH assay; DOPA oxidation inhibition; ferrous ions Chelating ability	100 μg/mL; 0.625, 12.5, 25, 50, 100μg/mL; 63-1000 μg/mL	[170]
25	Ethanol Extraction	Pregnant SD rats, male rats; Neuronal cells	i.g. 100 and 200 mg/kg; 10, 50, 100 μg/mL	[171]
26	Styrylpyrones	Fenton reaction	1 and 3 mM	[67]
27	Polysaccharides	Female Kunming mice; avtivities of catalase, SOD, GSH-PX, malondialdehyde MDA	i.g. 100, 200, 400 mg/kg BW	[172]
28	Protocatechualdehyde, protocatechuic acid, hispidin, caffeic acid, phelligridin D, gallic acid, 2,5-dihydroxybenzoic acid, ferulic acid, 2,3 dihydroxybenzaldehyde, arbutin, isoferulic acid, ellagic acid	Peroxynitrite ($ONOO^-$) scavening activity	Mean IC50=0.7~5.4 μM	[81]
29	Ethanolic extraction and hot water extraction	Reducing power effect; DPPH assay; ferrous ions chelating effect	EC50= 0.13~0.88; EC50=0.06~2.12; EC50=7.8~65.75	[173]
30	Methanol extraction	DPPH assay; effect on SOD and GSH-Px	12.5, 25, 50, 125, 250, 500 μg/mL	[174]
31	Methanol and hot water extractions	H2O2 and nitrite scavening activity	1, 5, 10 μg/mL	[175]
32	Ethyl acetate and *n*-BuOH extraction	Superoxide anion radical, hydroxyl radical, DPPH radical, NO radical scavening activities	5, 10, 20, 50, 100, 200 μg/mL; 0.05, 0.1, 0.5, 1, 5, 10 μg /mL; 5, 10, 20, 50, 80 μg /mL; 2, 10, 20 mg/mL	[176]

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33		Two flavonoid extracts	Reducing power and DPPH assay, ferrous ions Chelating ability; Hydroxyl free radical and superoxide anion scavening activites	EC50= 7.16, 4.62, 3.36, 3,47, 0.48 mg/mL	[177]
34		Polysaccharides	LPO, Reducing power, DPPH, Ferrous ions chelating assay	0.1-20 mg/mL; 0.1-20 mg/mL; 0.1–10 mg/mL; 0.1-20 mg/mL	[178]
35	Immuno-modulatory activity	Exopolysaccharides	Male BALB/c mice and male ICR mice; splenocyte and macrophages	25, 50, 100 μg/mL; 25, 50, 100, 200 μg/mL	[179]
36		Polysaccharides-protein complex	Murine (C57BL/6) spleens	10, 50, 100 μg/mL	[147]
37		Polysaccharides	T lymphocyte, Natural killer cells and macrophages from the spleens of BDFI mice; SD rat spleen cell	i.p. 100 mk/kg body weight; 0.01 and 0.1 mg/mL	[180]
38		Polysaccharides	Splenocytes from male inbred BALB/c mice; RAW 264.7 cells	0, 1, 10, 30 μg/mL; 0-100 μg/mL	[181]
39		Polysaccharides	Human peripheral blood mononuclear cells	50 μg/mL	[178]
40		A water-soluble polysaccharide	macrophage RAW 264.7	0, 100, 200, 400 μg/mL	[105]
41		Polysaccharides; aqueous and ethanol extractions	Human promonocytic cell line THP-1	1, 3, 5 μg/mL; 10, 30, 50, 100 μg/mL	[182]
42		Ethyl acetate extraction	Male ICR mice	2 mg/head/day	[183]
43		Water extraction	Male ICR mice	i.g. 0.01-1 g/kg	[184]
44		Ethyl acetate fraction	Male ICR mice	i.g. 2 mg/mouse	[185]
45		Crude polysaccharides	Splenocytes from male C-57 mice	50, 200, 500 μg/mL	[186]
46		Ethanol Extraction	Male SD rats	i.g. 200 mg/kg	[187]
47		Crude polysaccharide	Female C3H/He, BALB/c mouse	100 μg/mL	[188]
48		Mycelium Extract	Splenocytes from male SD rats	0.1, 1, 10 μg/mL	[148]

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49		Boiling water fraction	Male and female BALB/c and male ddY mice	i.g. 30, 100, 300 mg/kg	[16]
50		Polysaccharides	Male SD rats	i.g. 100, 200, 400 mg/kg	[189]
51		Water extraction	Male ICR mice; mesenteric lymph node lyphocytes	i.g. 2 mg/kg	[190]
52		Methanol fraction, water fraction and boiling water fraction	Female BALB/cCrSlc mice; U266B1 cells, primary B cells	painting 1 mg/each ear; 0.1, 0.25, 0.5 mg/mL; 0.25, 0.5, 0.75 mg/mL	[191]
53		Boiled water extracts	Female C57BL/6J mice	i.g. 1, 2, 4 g/kg/day	[192]
54		Water extraction	Male Balb/c, ICR mice and SD rats	i.g. 0.01-1 g/kg BW	[193]
55		Ethanol extraction	RAW 264.7 cells, *Brucella abortus* strain	0, 25, 50, 100 µg/mL; 0, 25, 50, 100, 200, 400 µg/mL	[146]
56		Water-ethanol extract	Female C57BL/6 mice	i.p. 100 mg/kg/day	[194]
57		Proteoglycan	Murine splenic lymphocytes from female C57BL/6 mice	0, 10, 50, 200, 500 µg/mL	[195]
58		Proteoglycan	Murine Bone Marrow (BM)-Derived Myeloid dendritic cells from male C57BL/6 and BALB/c mice	10, 50, 100 µg/mL	[15]
59	Hypoglycemic activity	Extracellular polysaccharides	Male SD rats	i.g. 200 mg/kg body weight	[196]
60		Extracellular polysaccharides	Male SD rats	i.g. 100 mg/kg	[197]
61		Polysaccharide	Male Kunming mice	i.g. 100 mg/kg/d body weight	[198]
62		Crude polysaccharides	Male Kunming mice	i.p. 100 mg/kg/day	[199]
63		*n*-hexane extracts	α-Glucosidase, α-Amylase	IC50= 0.165, 4.08 mg/mL	[200]
64		Water extraction	Male SD rats	i.g. 0.5 g/kg	[201]
65		Exopolysaccharides	Male C57BL/6J ob/ob mice	i.g. 200 mg/kg body weight	[202]
66		Crude polysaccharides	Male SD rats	i.g. 200 mg/kg body weight	[203]
67		Hispidin	RINm5F cells	30 µM	[168]
68		Polysaccharide	Female non-obese diabetic mice; ICR mice	i.p.30 mg/kg	[204]

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69		Extracellular polysaccharides	Male SD rats	i.g. 200 mg/kg body weight	[205]
70	Anti-inflammatory activity	*n*-BuOH subfraction	RAW264.7 macrophages	0, 0.5, 1, 2.5, 5 mg/mL	[206]
71		*n*-BuOH subfraction	RAW264.7 macrophages	0.1, 0.25, 0.5 mg/mL	[207]
72		Inotilone	Male ICR mice, RAW264.7 macrophages	i.g. 1.25, 2.5, 5.0 mg/kg; 0, 6.25, 12.5, 25 μM	[82]
73		Water extraction	Male SD rats	i.g. 20 mg/kg body weight	[208]
74		*n*-BuOH subfraction	RAW 264.7 macrophages	0, 25, 50, 100 μg/mL	[17]
75		Phellibaumins A, B, E, methylphelligrin A, *epi*-methylphelligrin A, methylphelligrin B, *epi*-methylphelligrin B	Prostate cancer cells (PC-3 cells)	IC50= 52.96, 41.40, 52.92, 36.44, 36.44, 22.46, 22.46 μM	[83]
76		Proteoglycan	Pathogen-free male DBA/1 inbred mice	i.g. 200 mg/kg	[209]
77		Ethanol and ethyl acetate extract	Female C57BL/6 mice	i.g. 200, 500 mg/kg/day	[210]
78		Aqueous Extracts	RAW 264.7 macrophages	10 μg/mL	[211]
79		Mycelium	Rat hepatocelluar carcinoma HepG2 cells	2.5, 5.0, 7.5 mg/mL	[212]
80		EtOH extract and *n*-BuOH subfraction	Male ICR mice	0.125, 0.25, 0.5, 1.0 mg/ear; i.g. 100mg/kg	[213]
81		Mycelium	Human hepatoma cells and Male SD rats	5 mL twice a day	[214]
82		Intracellular polysaccharides and exopolysaccharides	Male Kunming mice	i.g. 125, 250, 500 mg/kg	[215]
83		Aqueous extracts	Male SD rats	i.g. 10, 20 mg/kg	[216]
84		*P. linteus*-fermented broth; hispolon in PL fermented broth	RAW264.7 cells and murine primary peritoneal exudate macrophages	50, 100, 200, 400, μg/mL; 10 μg/mL	[217]
85	Anti-complement activity	Phellinusfurans A, B	*In vitro*	IC50= 33.6, 33.7 μM	[60]
86		Hot-water extraction	*In vitro*	1000 μg/mL	[218]

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87	Anti-obesity activity	Methanol, water and ethanol extracts	Lipase inhibitory activitiy	2 mg	[219-221]
88	Anti-hyperlipemia activity	Polysaccharides	Female SD mice	i.g. 100, 200 mg/kg/d	[222]
89	Anti-bacterial effect	*n*-BuOH fraction	Twelve methicillin-resistant *Staphylococcus aureus* (MRSA) stains	MIC= 63-125 μg/mL	[223]
90	Anti-mutagenicity activity	Various extracts	*Salmonella typhimurium* from male SD rats	0.02, 0.1, 0.5, and 1 mg	[224]
91	Anti-mutagenicity activity	Cultured broth, mycelia, and fruiting bodies	*Salmonella typhimurium* tester strain TA 98 and TA 100	0.02, 0.1 0.5, 1 mg	[225]
92	Induction maturation of murine dendritic cells	Acidic polysaccharides	Dendritic cells from male C57BL/6 and BALB/c mice	100 μg/mL	[226]
93	Anti-shock effect	Acidic polysaccharide	Female BALB/c mice	i.p. 100 mg/kg	[227]
94	Inducing prostate enlargement	Water extraction	Adult male Wistar Hannover strain rats	i.p. 1 mL	[228]
95	Anti-angiogenic activity	MeOH extract and other fractions	Human umbilical vein endothelial cells; ICR male mice	6.25, 12.5, 25, 50, 100 μg/mL; i.p. 50 μg/mL	[229]
96	Amination effect	β-glucan	Bronchoalveolar lavage cells	10, 20, 50, 100 μg/mL	[230]
97	Increase the Growth of *Caenorhabditis elegans*	Lipopolysaccharide and β-glucans	*Caenorhabditis elegans*	1, 10 μg/mL	[231]
98	Hypolipidemic activity	*Phellinus baumii*-fermented soybean	Male SD rats	i.g. Content 10% and 20%	[232]
99	Preventing the formation of gallstones	Fruiting body and mycelia powder	Male ICR mice	i.g. Content 3 %	[233]
100	Inhibiting the serotonin production	Aqueous extracts	Male SD rats	i.g. 50, 100, 200 mg/kg	[234]
101	Neuroprotective effect	Filtrate of *Phllinus linteus* broth culture	Male SD rats	i.g. various doses	[235]
102	Chempreventive activity	Methanol extraction	Hepalclc7 cells	0.1, 1, 2, 3 mg/mL	[236]

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103	Inhibition of cytochrome P450 isozymes	Polysaccharides	Liver microsomes from Male SD rats	0.5, 1, 3 mg/mL	[237]
104	Inhibition of cytochrome P450 isozymes and ornithine decarboxylase avtivities	Polysaccharides	Liver microsomes and mouse epidermal cells line 308 from Male SD rats	5, 10, 25, 50, 100 µg/mL	[238]
105	Inibitory of Aldose Reductase	Davallialactone, hypholomine B, ellagic acid	Crude aldose reductase from SD rats; Human Recombinant Aldose Reductase	IC50= 0.33, 0.82, 0.63 µM; IC50=0.56, 1.28, 1.37 µM	[84]
106	Antiviral activity	Hot water extract of mecylium	*Pepper mild mosaic virus* and *Cucumber mosaic virus*	1, 2, 3, 5, 10 mg/mL	[239]
107	Antioxidant activity	Methanolic extract	Lipoxygenase Inhibitory Activity	100 µg/mL	[240]
108	Neuraminidase inhibitory effect	Hypholomine B and hspidin	*In vitro*	0.01, 0.02, 0.06, 0.17 µM; 10, 20, 30, 40 µM	[86]
109	Neuraminidase inhibitory effect and antiviral activity	inotilone and 4-(3,4-dihydroxyphenyl)-3-buten-2-one	Recombinant influenza A virus H1N1; MDCK cells	IC50=29.1, 125.6 µM; IC50= 61.5, 52.3 µM	[87]
110	Anticoagulant activity	Mycelial extract	Activated partial thromboplastin time test	20 mg	[241]
111	Enoyl-ACP reductase inhibitory and antibacterial avtivities	Phellinstatin	*Staphylococcus aureus* FabI, *S. aureus* (MRSA) CCARM 3167	IC50= 6 µM; MIC= 128 lg/mL	[88]
112	Preventing tacrine-induced hepatotoxicity	Water extraction	HepG2 cells	0.0625, 0.125, and 0.25 mg/ml	[242]
113	Prolyl endopeptidase inhibitory activity	Inoscavin E and C	*In vitro*	Mean IC50= 4.26, 4.08 µM	[89]
114	Inhibiting protein glycation	Interfungins A	*In vitro*	0..2, 1, 2, 10 mM	[85]
115	Anti-asthmatic Effect	Ethanol extraction	Female BALB/c mice	i.g. 10 mg/kg	[243]

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116	Hematopoietive effect	Polysaccharide	Mouse splenocytes and bone marrow cells	500 µg/mL	[244]
117		Mycelial extract	Male SD rats	i.g. 5, 15 mg/kg	[245]
118		Water extraction	Male SD rats	i.g. 200 mg/kg	[246
119		Mycelial culture	Hepatocyte smale Wistar rats	100 mg/kg	[247]
120		Polysaccharide	Male SD rats	i.g. 50 mg/kg twice a day	[248]
121		Water extraction	Male SD rats	i.g. 5 g/kg/d	[249]
122	Inducing quinone reductase and glutathione S-transferase activity, and glutathione levels	Polysaccharides	Hepa lclc7 cells	0.1-1 mg	[151]
123	Increarse production of NO; Immunostimulatory	A water-soluble polysaccharide	RAW264.7 cells	0, 100, 200, 400 µg/mL	[105]
124	DPPH radical scavenging activity	Polysaccharides	DPPH	EC50= 420 µg/mL	[42]
125	Increarse production of NO	Polysaccharides	Bronchoalveolar lavage (BAL) cells	5, 10, 25, 50, and 100 µg/mL	[42]
126	β-secretase Inhibitory Activity	Polysaccharides	*In vitro*	1, 10,100 µg/mL	[39]
127	Anticomplementary activity	Hot water and alkali water extract	*In vitro*	50 µg/ml	[123]
128	NO prodution activity	Hot water and alkali water extract	Raw 264.7 murine macrophage cells	10 µg/ml	[123]
129	Lipid peroxidation activity	Hot water and alkali water extract	*In vitro*	1 mg/ml	[123]

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130	Inhibition of gap junctional intercellular communication through the inactivation of ERK1/2 and p38 MAP	30% ethanol extract	WB-F344 rat liver epithelial cells	5 and 25 µg/mL	[152]
131	Anti-angiogenic activity	70% ethanol extract	Chorioallantoic membrane (CAM) assay	2.5, 5, 10, 20 µg/egg	[13]
132	Decrease the Cholesterol	Water extraction	Male SD rats	i.g. 50 and 100 mg/kg	[167]
133	Melanogenesis Inhibitory	Methanol extraction	B16/F10 melanoma and NIH3T3 cells	0.625, 12.5, 25, 50, 100 µg/mL	[170]
134	Disease resistance of kelp grouper against *vibriosis*	Ethanol extraction	Kelp grouper, *E. bruneus*	10g/kg fed with diet	[250]
135	Anti-nociceptive activity	EtOH extract and *n*-BuOH subfraction	Male ICR mice	i.g. 100, 200, 400 mg/kg	[213]
136	Tyrosinase inhibitory activitiy	Protocatechualdehyde, 5-Hydroxymethyl-2-furaldehyd	*In vitro*	IC50= 0.4, 90.8 µg/mL	[55]

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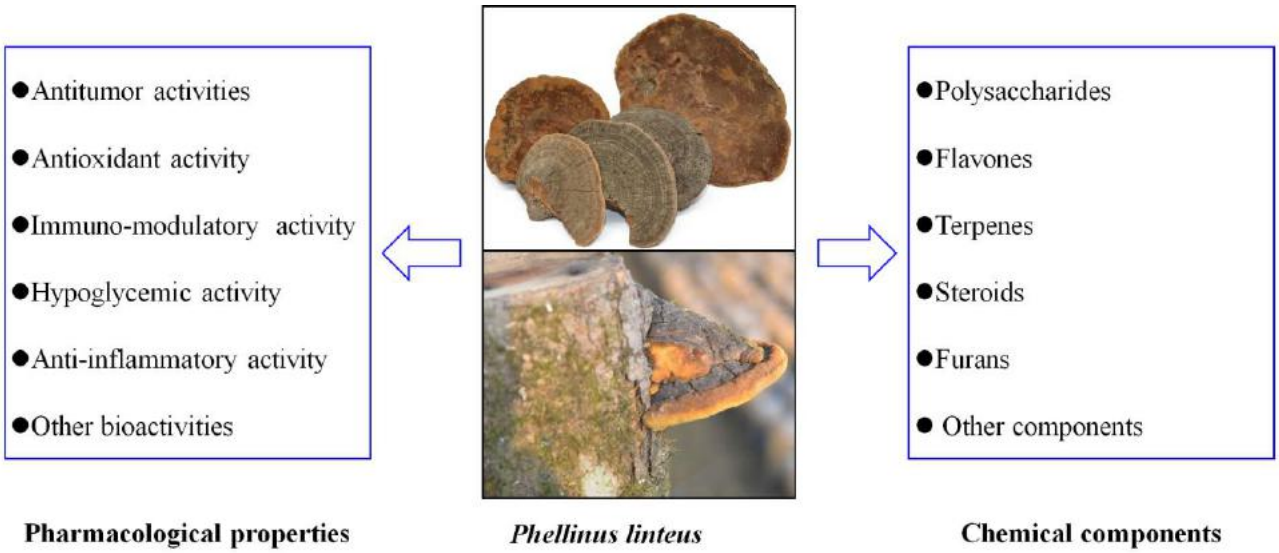
Graphical Abstract



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